

The Macroeconomic Consequences of Sovereign Credit Risk*

Preliminary – Comments welcome!

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Abstract

We propose a novel high-frequency identification strategy to isolate unanticipated changes in perceived sovereign credit risk for a large cross-section of economies. These surprises originate from sovereign credit rating announcements, are plausibly exogenous to fundamentals, and are not predictable using daily and monthly macroeconomic indicators. Yet they shift sovereign bond yields within two days and have considerable explanatory power for bond yields at lower frequencies. Our surprises combined with daily event-study methods show that increases in sovereign credit risk worsen stock prices and corporate financing conditions. Finally, using monthly local projections, we find that sovereign credit risk has adverse macroeconomic consequences, acting as a negative supply shock, which depresses economic activity but raises consumer prices.

Keywords: Sovereign credit risk, high-frequency identification, credit ratings

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1 Introduction

What are the macroeconomic consequences of a shock to sovereign credit risk? This question is studied by a large body of theoretical work, which, e.g., focuses on implications for fiscal and monetary policy (Uribe, 2006; Corsetti, Kuester, Meier, and Müller, 2013), and the effects on financial markets and the macroeconomy (Bocola, 2016; Farhi and Tirole, 2018; Thaler, 2021). Empirically, however, this question is difficult to answer because agents' risk assessments are shaped by macroeconomic conditions. The resulting reverse causality problem poses a significant identification challenge.

This paper addresses this challenge with a novel high-frequency identification approach. We isolate sovereign credit risk surprises in a large cross-section of economies using high-frequency changes in asset prices in a tight window surrounding sovereign credit rating announcements by private rating agencies. This identification approach is the first key contribution of our paper. The resulting surprises are plausibly exogenous to current fundamentals and are not predictable using daily and monthly macroeconomic indicators. Yet they shift sovereign bond yields within two days and have considerable explanatory power for bond yields at lower frequencies. As a second key contribution, we use our surprises to provide novel estimates on the consequences of sovereign credit risk on financial markets and the macroeconomy. First, using daily event-study methods, we show that increases in sovereign credit risk worsen stock market outcomes and corporate financing conditions. Second, using monthly local projections, we demonstrate that sovereign credit risk has adverse macroeconomic consequences, acting as a supply shock, which but raises consumer prices.

For our analysis, we collect sovereign credit rating announcements from the four largest private rating agencies. Our data is comprised of 53 economies between 1960 and 2026, with most countries entering the sample in the 1980s and 1990s. Based on these data, we compute a daily sovereign credit rating index for each country that captures the average creditworthiness of countries, as measured by sovereign credit ratings. The construction of this index is an additional contribution of our paper. We provide descriptive evidence on the

properties of this index and show that it is, as expected, inversely correlated with 10-year constant maturity government bond yields.

We combine our new sovereign rating index with daily 10-year sovereign bond yields and estimate parsimonious event studies that trace out the evolution of these yields on the days surrounding credit rating downgrades and credit rating upgrades. On average, there are no statistically significant pre-trends. Over the two days surrounding the rating change, however, bond yields shift significantly and persistently remain at the new level. This suggests that sovereign credit rating announcements are not fully anticipated by financial markets, a prerequisite for using these events for identification.

The event study evidence motivates us to use rating announcements to isolate sovereign credit risk surprises via high-frequency revisions in asset prices, a technique often applied to monetary policy announcements (e.g., [Kuttner, 2001](#); [Bernanke and Kuttner, 2005](#); [Nakamura and Steinsson, 2018a](#); [Altavilla, Brugnolini, Gürkaynak, Motto, and Ragusa, 2019](#); [Altavilla, Gürkaynak, Laeven, and Kind, 2025](#)). The key premise underlying this approach is that all relevant macroeconomic and financial information has been available before the rating change and has, therefore, already been priced in prior to the rating announcement.¹ Under this assumption, these surprises capture variation in sovereign credit risk that is plausibly exogenous to current macroeconomic fundamentals. This plausibly rules out reverse causality, i.e., that the measured changes in asset prices reflect shifts in macroeconomic and financial conditions rather than the credit rating announcement itself.²

We implement this identification approach as follows. First, we measure asset price revisions based on 10-year constant maturity sovereign bond yields at daily closing values. Using daily yields, as opposed to intraday data or futures, allows us to compute surprises for a large

¹We view it as implausible that rating agencies possess private information unavailable to other market participants, e.g., investment banks or hedge funds. Therefore, we expect that our credit rating surprises do not reveal any new fundamentals. This contrasts with central bank announcements (e.g., [Nakamura and Steinsson, 2018a](#); [Miranda-Agrippino and Ricco, 2021](#); [Bauer and Swanson, 2023a](#)), where it is more conceivable that these institutions possess superior information.

²These announcements are still likely to affect yields because they can, for example, coordinate investor expectations and trigger mandatory portfolio rebalancing by institutional investors.

cross-section of countries.³ Second, we choose an event window of two days to give markets sufficient time to process the underlying credit rating change. Importantly, this choice is in line with our previously discussed event study evidence, according to which yields continuously increase over this two-day window. Lastly, we remove rating announcements where relevant macroeconomic information is released within the event window, and rating changes where the yield data displays anomalies that are likely related to low trading volumes in the securities underlying the constant-maturity sovereign bond yields. We term our resulting series of yield changes around sovereign credit rating events *yield change surprises*.

In total, we obtain more than 1200 rating events across 51 economies. This broad coverage is a key advantage of our identification approach, allowing us to study sovereign credit risk in a large cross-section of economies. We provide several diagnostics for our newly constructed yield change surprises. First, corroborating the validity of the surprises, we document the absence of pre-trends with event study methods and that the surprises are not predictable by past daily and monthly macroeconomic data. Second, focusing on explanatory power, we document that the measured effect on yields is highly persistent over the days following a credit rating announcement. In a similar vein, we estimate a stylized first stage by projecting monthly sovereign bond yield changes on the surprises, which are summed to a monthly frequency. The estimates are highly statistically significant and produce (robust) Kleibergen-Paap F statistics well above 20 for most specifications. Overall, the diagnostics suggest that our surprises have meaningful explanatory power for sovereign bond yields and are likely unrelated to macroeconomic fundamentals. Thus, we proceed to use these surprises to assess the effect of sovereign credit risk on asset prices and macroeconomic outcomes.

To study the effect of sovereign credit risk on asset prices, we estimate macro event studies similar to [Kuttner \(2001\)](#) and [Gürkaynak, Kısacikoğlu, and Wright \(2020\)](#). We do so by projecting changes in asset prices in a two-day window around credit rating announcements

³Beyond data availability, information leakage in the hours leading up to a rating change is plausibly a more important concern for private companies (compared to central banks), which we may mitigate by using daily instead of intraday data. Using daily data is also not uncommon to isolate high-frequency surprises (e.g., [Känzig, 2021](#)).

on the associated yield change surprises. Based on this approach, we find a statistically significant response of country-specific stock price indices, although at relatively modest magnitudes. Specifically, a one-standard-deviation surprise (about 10 basis points) yields a decline in stock prices of one-tenth of a standard deviation. We further document meaningful heterogeneity depending on the sign of the surprise as well as cross-country heterogeneity. This exercise suggests that financial markets anticipate that sovereign credit risk is contractionary, albeit with small magnitudes. Additionally, we use the same methodology to study corporate bond prices and find that they decline more strongly and significantly. This result suggests that private funding costs may be an important channel through which sovereign credit risk transmits to the real economy, consistent with [Bocola \(2016\)](#) and [Arellano, Bai, and Bocola \(2026\)](#).

As a final step of our analysis, we turn to the medium-term macroeconomic consequences of sovereign credit risk. To this end, we estimate monthly local projections, modeling the high-frequency surprise as a shock to sovereign credit risk. The resulting OLS estimates are highly statistically significant. The estimates reveal that the effects on sovereign bond yields are very persistent over the three-year period under consideration. Stock markets are initially somewhat unresponsive, consistent with modest effects in the macro event study. After a few months, however, stock markets have deteriorated more strongly, suggesting that financial market participants may initially underestimate the impact of sovereign credit risk. Consistent with weaker stock markets, we find a pronounced decline in industrial production and an increase in unemployment, confirming that sovereign credit risk has real adverse consequences. Lastly, and contrary to monetary policy interest rate changes, we find that sovereign credit risk raises consumer prices.

Taken together, these results indicate that sovereign credit risk acts as a negative supply shock. The increase in consumer prices may arise via a cost channel since sovereign credit risk raises corporate funding costs, as is also shown using the macro event study. The price increase could also partly reflect a fiscal dominance channel where investors discount future

primary surpluses more strongly in response to higher credit risk (e.g., [Leeper, 1991](#); [Sims, 1994](#); [Cochrane, 2001](#)). In this case, the intertemporal government budget constraint may dictate a higher equilibrium price level when the fiscal corrections are insufficient. Lastly, given that the effects on sovereign bond yields are highly persistent, the increase in consumer prices may also partly operate via a Fisher channel (e.g., [Mishkin, 1992](#); [Uribe, 2022](#)). Here, persistently higher nominal rates induce higher inflation when equilibrium conditions require a lower real rate. The decline in real activity, in turn, may be partly demand driven, e.g., via less government spending and less private investment due to the documented increase in funding cost, and potentially elevated uncertainty.

Related literature. We contribute to two broad strands of the literature. First, we contribute to the literature on high-frequency identification specifically and causal identification in macroeconomics more broadly; see [Ramey \(2016\)](#) and [Nakamura and Steinsson \(2018b\)](#) for an overview. Related are papers that apply a high-frequency identification approach in a wide range of contexts. Most often, this approach is applied to monetary policy (e.g., [Kuttner, 2001](#); [Bernanke and Kuttner, 2005](#); [Nakamura and Steinsson, 2018a](#)), and has been refined to better account for potential confounders (e.g., [Miranda-Agrippino and Ricco, 2021](#); [Bauer and Swanson, 2023a,b](#)). Beyond these U.S.-focused papers, [Altavilla et al. \(2019\)](#) and [Braun, Miranda-Agrippino, and Saha \(2025\)](#) apply this type of identification approach to the Euro Area and the U.K., respectively. Recently, several papers measure high-frequency surprises due to speeches of central bankers (e.g., [Swanson, 2023](#); [Swanson and Jayawickrema, 2023](#); [Altavilla et al., 2025](#); [Odendahl, Istrefi, and Sestieri, 2025](#)). These events are more similar to our credit rating changes because they are scheduled in a more discretionary manner.

Beyond monetary policy, the high-frequency approach has been applied to OPEC oil quota announcements ([Känzig, 2021](#); [Degasperi, 2023](#)), European carbon pricing events ([Känzig, 2023](#)), the Georgia Senate election runoff ([Hazell and Hobler, 2024](#)), and IMF announcements

and court decisions in Argentina ([Hébert and Schreger, 2017](#); [Sagawa, 2025](#)). Further closely related are papers that identify government debt supply shocks from debt auctions for the U.S. ([Phillot, 2025](#); [Bi, Phillot, and Zubairy, 2025](#)) and the U.K. ([Joyce and Lengyel, 2024](#)). These papers deliver debt supply shocks, whereas our surprises capture variation in perceived sovereign credit risk. Finally, [Bahaj \(2020\)](#) applies a high-frequency identification strategy to Euro crisis events. The author identifies shocks to sovereign credit spreads focusing on country-specific policy events rather than rating changes. Our approach, instead, does not rely on a particular crisis, like the Euro crisis, to generate variation in (perceived) risk, but allows the identification of broader patterns in a large cross-section of advanced and emerging economies. More generally, none of the above papers identifies the effects of sovereign credit risk using rating announcements. Proposing and implementing this novel identification approach is the principal contribution of this paper.

Further complementary are several works that seek to understand credit ratings, focusing on implications for GDP growth ([Chen, Chen, Chang, and Yang, 2016](#)), home bias ([Fuchs and Gehring, 2017](#)), corporate ratings ([Almeida, Cunha, Ferreira, and Restrepo, 2017](#)), rating quality and competition ([Becker and Milbourn, 2011](#)), as well as cross-country spillovers ([Gande and Parsley, 2005](#)). A comprehensive survey is provided by [Panizza, Sturzenegger, and Zettelmeyer \(2009\)](#).

The second strand of literature is concerned with the object of interest that we seek to identify: sovereign credit risk. A large theoretical literature studies sovereign credit and default risk (e.g., [Arellano, 2008](#); [Aguar and Gopinath, 2006](#); [Mendoza and Yue, 2012](#)). Further related papers focus on the relation between sovereign default risk and private financing costs (e.g., [Gennaioli, Martin, and Rossi, 2014](#); [Bocola, 2016](#); [Arellano et al., 2026](#)) and the feedback between sovereign and banking sector fragility, including the possibility of sovereign-bank doom loops (e.g., [Farhi and Tirole, 2018](#); [Sosa-Padilla, 2018](#); [Thaler, 2021](#)). Relative to these papers, we provide direct empirical evidence on the consequences of sovereign credit risk without assuming a fully specified structural model. In line with theoretical works, we

find that sovereign credit risk is contractionary and partly operates through higher corporate funding costs.

Empirically, a host of papers studies sovereign risk, stress, and crisis episodes, the transmission of sovereign risk to banks and credit markets, and the Euro Area sovereign debt crisis in particular (e.g., [Cruces and Trebesch, 2013](#); [Acharya, Drechsler, and Schnabl, 2014](#); [Altavilla, Pagano, and Simonelli, 2017](#); [Brutti and Sauré, 2016](#); [Ongena, Popov, and Van Horen, 2019](#)). These papers often exploit cross-sectional variation in bank exposures or country-specific crisis episodes. We contribute to this literature by constructing and exploiting surprise shifts in perceived sovereign credit risk for a large cross-section of economies spanning multiple decades. This allows us to estimate the macroeconomic consequences of sovereign credit risk beyond a specific crisis episode, which is the second key contribution of this paper. Directly related is also [Diebold and Hack \(2026\)](#), which applies our identification approach to study the composition of sovereign debt, highlighting the role of loans to the government in response to yield changes.

2 Data

In this section, we first introduce all data, covering our newly constructed sovereign credit rating index as well as all financial and macroeconomic data used in our subsequent analysis. An overview of our sample coverage by data type is given in [Table A.1](#) and in [Table A.2](#) we show the corresponding summary statistics. We then offer a first look at the data on credit rating announcements and examine the correlation between our rating index and sovereign bond yields.

2.1 Financial and macroeconomic data

We use ratings, macroeconomic, and financial data at monthly and daily frequency.

We obtain 10-year constant maturity sovereign bond yields at daily closing values, covering

51 countries between 1960 and 2025. From these, we also construct monthly bond yields by taking end-of-month values. At the intraday level, we obtain the same variable at 5-minute frequency for 49 countries between 2021 and 2026.⁴ Further financial data at daily frequency are the closing prices of country-specific main stock indices, covering 50 economies, and a proxy for corporate bond prices (based on indices or corporate bond ETFs, depending on data availability for each country), covering 17 economies since the late 1990s and early 2000s.

Lastly, we also use monthly macroeconomic data on industrial production, unemployment, and price indices for 27 economies in our sample; for details of variable-level coverage see [Table A.1](#).

2.2 Credit ratings

We collect the majority of sovereign credit ratings from [World Government Bonds Credit Ratings \(2026\)](#), and, in the case of missing entries, supplement them using [Trading Economics \(2026\)](#) and the websites of [Fitchs \(2026\)](#); [Moody's \(2026\)](#); [Standard and Poors \(2026\)](#); [DBRS \(2026\)](#).⁵ A commonality among the rating schemes of all four agencies is that they offer the same number of distinct rating categories.⁶ This similarity offers a natural way of aggregating all four ratings into a compound rating index, since all ratings can be ranked on a numerical scale from 0 to 32. [Figure A.1](#) in the Appendix exemplifies this for the United States, showing its ratings and the complete set of different scales of the four agencies as of May 2026.

To obtain a composite rating index, we map the ratings on a numerical scale for each rating agency. Ratings for governments in default are coded as 0. The lowest non-default rating is

⁴Note that for countries with small or illiquid bond markets, these intraday series may frequently remain unchanged for several consecutive observations due to insufficiently low trading volumes of the underlying bonds.

⁵We include these rating agencies as the only ones classified as External Credit Assessment Institutions (ECAI) by the EBA ([European Banking Authority, 2018](#)), coinciding with them having the largest market shares of all rating agencies.

⁶Labels may differ across agencies to, e.g., reflect different corporate styles. Yet, the individual labels are remarkably close: for example, the current (end of May 2026) US rating of S&P, Moody's, and Fitch is the second-best rating on their individual rating scales, with labels "AA+", "Aa1", and "AA+", respectively.

coded as 1, which, e.g., in the case of S&P corresponds to “C” with a negative outlook. It increases to 1.5 for a stable outlook and to 2 for “C” with a positive outlook. Afterwards, each rating step has three potential sub-steps: negative, stable, and positive outlook, with each improvement increasing our rating variable by 0.5. This implies that the top rating is coded as 32.⁷ The other rating agencies use comparable scales, as illustrated in [Figure A.1](#). Employing this rating-coding approach, we obtain a daily rating time series for each country and rating agency. Rating announcements on weekends are coded to the subsequent Monday, so that our rating series is at the trading-day frequency.

Given these data, we compute our sovereign credit rating index as the cross-sectional average across all four rating agencies for every country-time observation. In total, our rating index covers 53 economies between 1960 and 2026, with most countries entering the sample in the 1980s and 1990s.

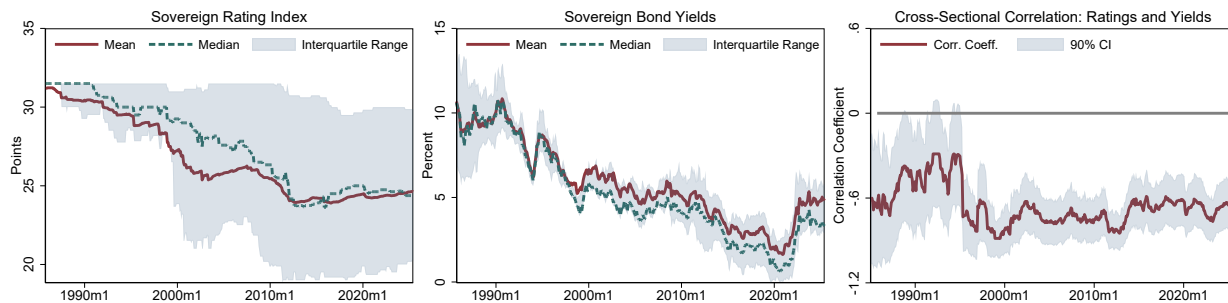
2.3 Sovereign credit ratings and bonds yields over time

To understand the variation in our sovereign credit rating index over time, we provide descriptive evidence and a comparison with sovereign bond yields. We do so in [Figure 1](#), aggregating both variables to a monthly frequency to smooth out short-term variability.

The left panel of [Figure 1](#) displays the mean, median, and interquartile range of our rating index between 1985 and 2025. The figure reveals a secular decline in sovereign credit ratings, visible in both the median and the average. This trend is partly driven by the inclusion of additional countries with lower credit scores over time, and partly by downgrades within the existing sample, particularly around the 2001 and 2008 financial crises. Notably, these rating deteriorations are concentrated among the lowest-rated sovereigns: the upper bound of the interquartile range — the third quartile — remains relatively stable throughout the sample period.

⁷The different rating steps for, e.g., S&P are: C, CC, CCC-, CCC, CCC+, B-, B, B+, BB-, BB, BB+, BBB-, BBB, BBB+, A-, A, A+, AA-, AA, AA+, and AAA. [Figure A.1](#) illustrates this for S&P and the rating steps of the other agencies.

Figure 1: Sovereign credit ratings and government bond yields



Notes: This figure displays monthly sovereign credit ratings and 10-year government bond yields. The left panel displays the arithmetic average and median of our credit rating index as solid red and dashed green lines, respectively. The shaded area represents the interquartile range of the rating index. The middle panel presents the same three statistics, mean, median, and interquartile range, for bond yields for the same set of observations. The right panel displays the time series of cross-sectional correlation coefficients between bond yields and the rating index, and the shaded area indicates the 90% confidence set.

The middle panel displays the 10-year government bond yield for the same set of observations, i.e., observations for which both ratings and yields are available. Despite notable short-term fluctuations, yields exhibit a similar long-term downward trend. This secular trend in bond yields aligns well with a large literature documenting long-term declines in interest rates (Holston, Laubach, and Williams, 2017; Rogoff, Rossi, and Schmelzing, 2024), particularly for safe assets (Caballero, Farhi, and Gourinchas, 2017; Del Negro, Giannone, Giannoni, and Tambalotti, 2019). As might be expected, this trend partially halts around the 2008 Financial Crisis and the subsequent Euro Crisis, but resumes afterward.

The observed positive co-movement between average credit ratings and average bond yields likely reflects broader macroeconomic trends that drive down bond yields regardless of credit rating developments. For example, this may be due to an increased global supply of savings, often referred to as a savings glut (Bernanke, 2005; Summers, 2014; Mian, Sufi, and Straub, 2020). To measure the relationship between yields and credit ratings independent of such secular trends, we compute the monthly cross-sectional correlation between sovereign bond yields and credit ratings over time. The results are displayed in the right panel, indicating that ratings and yields are inversely correlated, as expected.

3 Identification of bond yield surprises

In this section, we use our newly constructed sovereign credit rating index to identify changes in sovereign credit risk that are plausibly exogenous to macroeconomic fundamentals. To do so, we proceed in three steps. First, we estimate parsimonious event studies that document how sovereign bond yields evolve on trading days around credit rating announcements. We then discuss the identification challenge posed by reverse causality and explain how our high-frequency surprises are constructed to address it. Finally, we present a range of diagnostics showing that our surprises are not predictable by macroeconomic indicators and that they have meaningful explanatory power for sovereign bond yields.

3.1 A first look at sovereign credit rating announcements

Event study specification. To understand how sovereign bond yields evolve around sovereign credit rating changes, we begin by estimating the following daily event study regression

$$Y_{i,\tau+h} - Y_{i,\tau-1} = \alpha_i^h + \beta^h \text{RatingChange}_{i,\tau}^k + u_\tau^h, \quad (1)$$

for $h \in [-10, 10]$, where $Y_{i,\tau}$ is the daily 10-year sovereign bond yield at closing value, α_i^h is a country fixed effect, and u_τ^h is an error term. The main regressor of interest, $\text{RatingChange}_{i,\tau}^k \in \{0, 1\}$, is a credit rating announcement indicator, where $k \in \{\mathcal{D}, \mathcal{U}\}$ refers to credit rating downgrades and upgrades, respectively.⁸

The distinction between credit rating downgrades and upgrades allows us to remain agnostic about the exact measurement of the potentially underlying sovereign credit risk shock. However, this comes at the expense of potentially noisy estimates, as our indicator variable does not account for the fact that rating changes can be fully or partially anticipated.

⁸To be precise, we have $\text{RatingChange}_{i,\tau}^{\mathcal{U}} = \mathbb{I}\{\text{RatingIndex}_{i,\tau} > 0\}$ and $\text{RatingChange}_{i,\tau}^{\mathcal{D}} = \mathbb{I}\{\text{RatingIndex}_{i,\tau} < 0\}$, where $\mathbb{I}\{\cdot\}$ is an indicator function and $\text{RatingIndex}_{i,\tau}$ is the credit rating index of country i at trading day τ , as introduced in [Section 2.2](#).

In fact, even the sign of the yield response is not clear from an ex-ante perspective. For example, if markets expect a full downgrade but the rating agency only shifts to a negative outlook, this could be interpreted as “good news”, with yields falling in response to the downgrade. Nevertheless, we find this parsimonious specification to be informative about the degree of anticipation of rating changes. Lastly, to ensure our results are not overly sensitive to extreme outlying observations, we winsorize the left-hand side variable at $\pm 0.5\%$.⁹

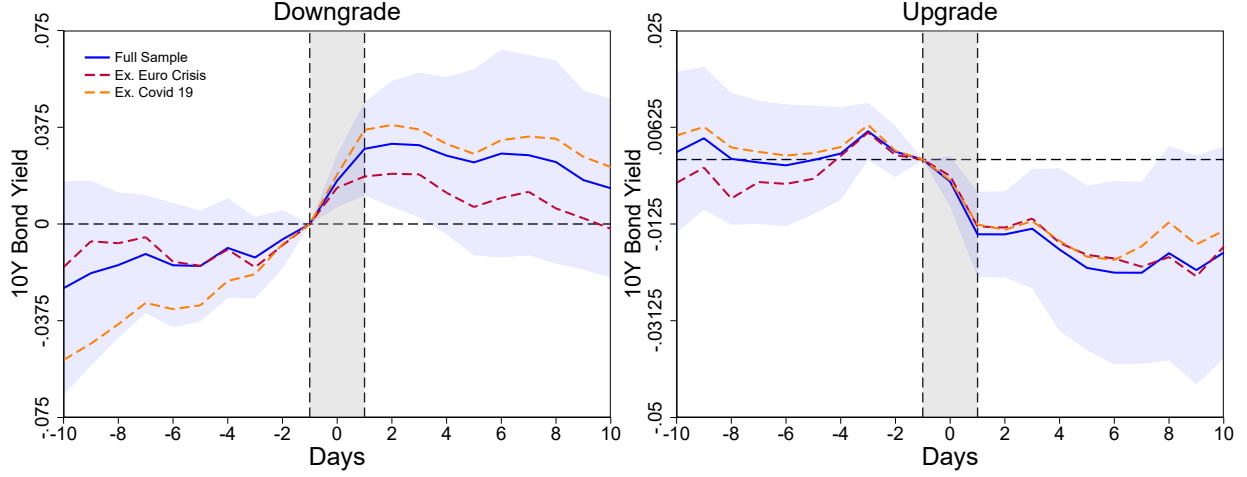
Event study results. The resulting event study point estimates are shown as solid blue lines in [Figure 2](#) for downgrades (left panel) and upgrades (right panel), respectively. The blue-shaded area indicates the corresponding 95% confidence bands, based on standard errors clustered at the country and time level. The red and orange dashed lines show the point estimates when excluding either the Euro Crisis or Covid-19 observations, respectively.

For both rating downgrades and upgrades, we do not detect any statistically significant pre-trends. However, as expected, the associated confidence bands are relatively wide, suggesting meaningful heterogeneity across rating announcements, presumably related to the anticipation of events. This reinforces the need for a high-frequency identification strategy, which differences out potentially heterogeneous pre-announcement dynamics.

Post announcement, we find a statistically significant change in yields. This jump in yields indicates that, on average, sovereign credit rating announcements are not fully anticipated and provide potentially useful identifying variation in sovereign credit risk as perceived by financial markets. Interestingly, in both cases, the yield change takes precisely one day after the announcement to materialize. To illustrate this, we also display a two-day event window from $h = -1$ to $h = 1$ as a grey-shaded area. We will return to this finding when constructing

⁹To be precise, we set values below (above) the 0.5 percentile (99.5 percentile) to the value of that percentile. For completeness, in [Figure B.1](#) in [Appendix B](#), we show how the estimates change when not winsorizing the data or when doing so at $\pm 1\%$. The comparison reveals that estimates are confounded by very few extreme events when winsorized. Often, this reflects the fact that yield data may become very volatile when trading volumes of the underlying securities are low, which may be the case for small or developing economies with limited sovereign debt markets. Conversely, winsorizing at $\pm 1\%$ leaves the results relatively unchanged. If anything, the pre-trend estimates move closer to zero and the post-announcement estimates become more stable, both suggesting that our baseline treatment of outliers is conservative.

Figure 2: Sovereign bond yields around credit rating announcements



Notes: The figure displays event study OLS estimates based on Equation 1 at trading day frequency, as specified in Section 3.1. The left-hand side variable is the 10-year sovereign bond yield, and the estimates correspond to an indicator variable that is activated on days of sovereign credit rating downgrades (left panel) and upgrades (right panel), respectively. The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels. The dashed lines report the point estimates of the same specification, when dropping all observations in years 2010 to 2013 (Ex. Euro Crisis) and in years 2020 to 2023 (Ex. Covid 19).

the high-frequency yield surprise.

Overall, we find that sovereign credit rating announcements are followed by statistically significant shifts in bond yields, and there is limited evidence of pre-trends on average. Together, these findings imply that credit rating announcements are not fully anticipated by financial markets. Finally, these patterns are a robust feature of the data and are not driven by the Euro Crisis or Covid-19 observations. We therefore conclude that sovereign credit rating announcements provide valuable information on changes in sovereign credit risk as reflected in bond yields, and may be used to identify sovereign credit risk surprises via a high-frequency identification strategy.

3.2 Constructing yield change surprises

We now briefly discuss the identification challenge posed by reverse causality and explain how our high-frequency surprises are constructed to address this identification concern.

Identification approach. The key identification idea is that high-frequency sovereign bond yield changes surrounding credit rating announcements can be used as a source of variation that is plausibly exogenous to macroeconomic fundamentals.¹⁰ The underlying premise is that all relevant macroeconomic and financial information has been available before the rating change and has, therefore, already been priced in prior to the rating announcement. This plausibly rules out reverse causality, i.e., that the measured yield changes reflect shifts in macroeconomic and financial conditions rather than the credit rating announcement itself.¹¹ One may wonder why rating changes should still move markets if all relevant information is already incorporated in asset prices. We argue that rating changes still matter for several reasons. First, they may coordinate market expectations around a widely observed signal. Second, institutional investors are often subject to regulatory or contractual constraints that restrict their holdings to assets above a certain rating threshold. As a result, investors may be compelled to rebalance their portfolios following a rating change, independently of any new fundamental information. Thus, bond yield changes in a sufficiently tight window around credit rating announcements reflect a surprise shift in financial markets’ assessment of sovereign credit risk that is plausibly exogenous to current macroeconomic fundamentals.¹²

High-frequency yield data. In the context of monetary policy, previous studies use prices of futures contracts and intraday data to explicitly capture expectation revisions for high-frequency identification (e.g., [Bernanke and Kuttner, 2005](#); [Gürkaynak et al., 2005](#)). However, this is infeasible in our setting since such futures and intraday data are not available for sovereign bond yields in our large cross-section of countries going back beyond the 1980s.

¹⁰This identification approach has been originally proposed to study monetary policy (e.g., [Kuttner, 2001](#); [Bernanke and Kuttner, 2005](#); [Gürkaynak, Sack, and Swanson, 2005](#)).

¹¹We view it as implausible that rating agencies possess superior private information unavailable to other market participants, such as investment banks or hedge funds. We therefore expect that information effects, as discussed in the context of monetary policy (e.g., [Nakamura and Steinsson, 2018a](#)) and OPEC oil supply shocks (e.g., [Degasperi, 2023](#)), are unlikely to be a concern here.

¹²Sovereign credit rating changes may also affect the credit conditions of private firms via the sovereign yield floor, i.e., the rate at which the sovereign itself borrows setting a lower bound on domestic borrowing costs. Changes in private credit conditions may therefore represent an important channel through which sovereign credit risk propagates to the broader economy, which we aim to capture with our surprises and discuss further later in the paper.

Instead, we operationalize our high-frequency identification approach using daily ten-year constant maturity government bond yields. With these data, we can measure bond yield adjustments in a tight event window around credit rating announcements. Importantly, daily bond yield changes remain adequate for constructing a yield surprise, as long as they correlate with the underlying fundamental shock to sovereign credit risk that materializes at the time of the credit rating announcement.

Event window choice. When constructing high-frequency surprises, the choice of the event window is critical. A narrower window reduces the risk of confounding events being inadvertently captured in the surprise variable, the so-called background noise (Nakamura and Steinsson, 2018a). A wider event window, on the other hand, helps to measure the effect of the rating change more comprehensively if market participants need more time to process the news and adjust their trading behavior following the announcement. A related concern is the potential for information leakage from rating agencies, which is likely more pronounced than for central banks, where announcements are typically made immediately after the policy decision is taken.¹³ Thus, an overly tight window may prevent us from measuring the underlying surprise adequately.¹⁴ Balancing these considerations, we opt for a two-day event window, which is also supported by the event study evidence in Figure 2. Based on this reasoning, we formally define a yield change surprise $YCS_{i,\tau}$ for country i on trading day τ as

$$YCS_{i,\tau} = (Y_{i,\tau+1} - Y_{i,\tau-1}) \times RatingChange_{i,\tau}, \quad (2)$$

where $RatingChange_{i,\tau} \in \{0, 1\}$ is an indicator variable that equals one on days on which the credit rating index for country i changes and zero otherwise.

¹³In contrast, for private non-policy institutions such as rating agencies, leakage is not only more likely, but it is also unclear with what lag relative to the internal decision the rating change is publicly announced.

¹⁴In addition, many rating changes are announced in the evening after markets have closed, or on weekends, to avoid undue market turmoil.

Potentially confounding events. We exclude events (i.e., set the corresponding surprises to zero) in two cases. First, we remove rating changes for which other relevant macroeconomic information is revealed within the event window, as is common practice to reduce background noise and sharpen identification. To do so, we assemble a database of potentially confounding events and exclude rating changes for which we identify a confounder within the event window. The potential confounders included are monetary policy announcements, announcements of sovereign haircuts and debt restructurings, announcements of official lending from the IMF and European institutions, and political events such as relevant referendums and government collapses. For sovereign haircuts and debt restructurings, coverage is only available at a monthly frequency, so we exclude all rating changes within the affected month. Throughout, we focus on country-specific confounders.¹⁵ Second, we remove events for which there are anomalies in the yield data. Specifically, we exclude the observations for which we winsorized bond yields within our event window when estimating the event studies in [Figure 2](#).

Heterogeneity in yield change surprises. Below, we further decompose the yield change surprise to capture heterogeneity in market responses, beginning by distinguishing between downgrades and upgrades. As previously discussed, the sign of the yield change associated with an upgrade or downgrade is not clear a priori due to the role of market expectations. In addition to this distinction, we therefore compile two further series capturing positive and negative yield change surprises separately. Formally, these four surprise series

¹⁵For example, a monetary policy announcement by the Bank of England does not constitute a confounder for Spanish credit rating announcements.

are given by

$$YCS_{i,\tau}^{\mathcal{D}} = YCS_{i,\tau} \times \mathbb{I}\{\Delta RatingIndex_{i,\tau} < 0\}, \quad (3)$$

$$YCS_{i,\tau}^{\mathcal{U}} = YCS_{i,\tau} \times \mathbb{I}\{\Delta RatingIndex_{i,\tau} > 0\}, \quad (4)$$

$$YCS_{i,\tau}^{\mathcal{H}} = YCS_{i,\tau} \times \mathbb{I}\{YCS_{i,\tau} > 0\}, \quad (5)$$

$$YCS_{i,\tau}^{\mathcal{L}} = YCS_{i,\tau} \times \mathbb{I}\{YCS_{i,\tau} < 0\}, \quad (6)$$

where $\mathcal{D}$ and $\mathcal{U}$ denote downgrades and upgrades, and $\mathcal{H}$ and $\mathcal{L}$ denote surprises associated with higher and lower yields, respectively.

Aggregation. In the following section, we study the yield change surprises at the daily and at the monthly frequency. To obtain surprises at the monthly frequency, we sum up all surprises within a month for each country in our sample. In [Appendix B](#), we also show that in the majority of months, this aggregation is meaningless because there is only a single rating change, see [Figure B.2](#).

3.3 Diagnostics for yield change surprises

This section presents summary statistics of the surprises, revisits the event study evidence from [Section 3.1](#), and provides a range of diagnostics for our surprises. The results show (i) that our surprises are not predictable by macroeconomic indicators, and (ii) that the surprises have meaningful explanatory power for sovereign bond yields.

Summary statistics. In [Table 1](#), we provide summary statistics for each of the five daily surprise series.¹⁶ The final column shows the number of countries for which we observe at least one non-zero surprise. In total, we have 1262 surprises, spread across upgrades and downgrades as well as positive and negative yield changes. Splitting by the direction of the rating change, we have 509 downgrades and 753 upgrades. Splitting by the sign of the

¹⁶We further report the same summary statistics after aggregation to monthly frequency in [Table B.1](#).

Table 1: Summary statistics for daily yield change surprises

	Mean	Median	Std. Dev.	Min	Max	Obs	Countries
$YCS_{i,t}$	-0.001	-0.002	0.105	-0.442	0.450	1262	51
$YCS_{i,t}^{\mathcal{D}}$	0.020	0.012	0.111	-0.442	0.450	509	47
$YCS_{i,t}^{\mathcal{U}}$	-0.015	-0.010	0.099	-0.415	0.358	753	51
$YCS_{i,t}^{\mathcal{H}}$	0.074	0.050	0.075	0.000	0.450	611	49
$YCS_{i,t}^{\mathcal{L}}$	-0.072	-0.047	0.077	-0.442	-0.001	651	51

Notes: This table presents summary statistics for all daily yield change surprises, as described in the text. The last column indicates the number of countries for which we have at least one non-zero surprise in the sample.

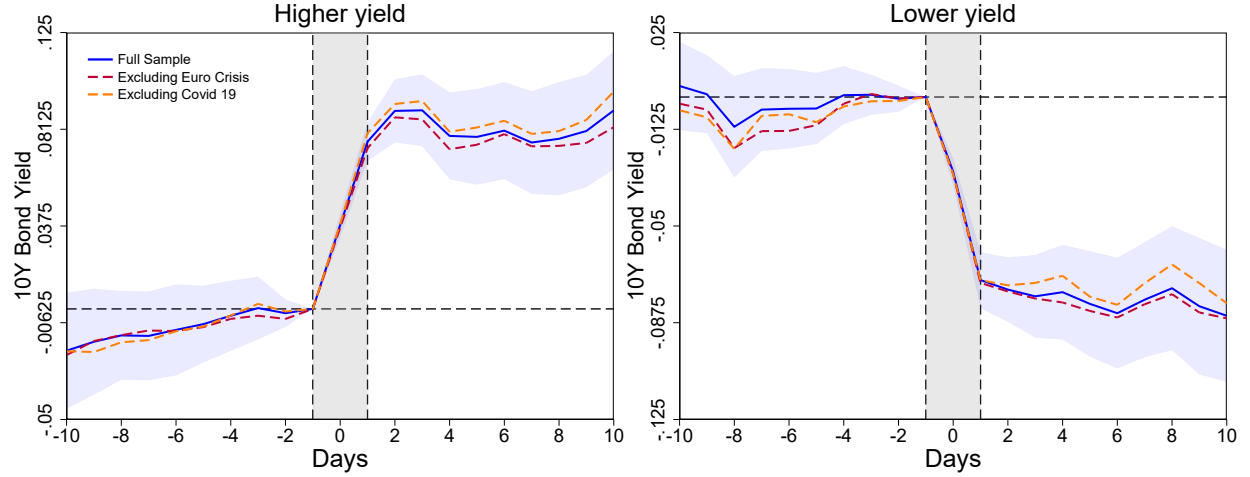
surprise, we have 611 positive and 651 negative surprises in our sample. Relative to high-frequency monetary policy surprises, our surprises are of comparatively large magnitude, with the standard deviation ranging between 0.08 and 0.11 percentage points and the largest surprises being around half a percentage point in absolute value. This suggests that our surprise series captures economically meaningful variation in sovereign credit risk.

Event studies. To better understand our yield change surprises, we revisit the event study design from [Figure 2](#). This is useful to assess whether our surprise events display pre-trends and whether they deliver persistent shifts in sovereign bond yields.

To do so, we estimate the same regression as in [Equation 1](#), but replace the indicator variable of interest with a new indicator based on our surprise series. Specifically, we now use an indicator that is activated only on days with a positive yield surprise, or alternatively, only on days with a negative yield surprise. The results are shown in [Figure 3](#) for the full sample and for subsamples excluding the Euro Crisis or Covid-19 observations.

For both cases, we find no statistically significant pre-trend, with point estimates remaining close to zero throughout the pre-announcement window. By construction, yields shift in the expected direction following the surprise within the event window. Importantly, however, yields remain persistently elevated at the new level after the announcement. This is *not* by construction but rather shows that our surprises capture changes in sovereign credit risk that persistently shift bond yields.

Figure 3: Sovereign bond yields around rating change surprises



Notes: The figure displays event study OLS estimates based on Equation 1 at trading day frequency, as specified in Section 3.1. The left-hand side variable is the 10-year sovereign bond yield, and the estimates correspond to an indicator variable that is activated on days with positive (left panel) and negative yield change surprises (right panel), respectively. The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels. The dashed lines report the point estimates of the same specification, when dropping all observations in years 2010 to 2013 (Ex. Euro Crisis) and in years 2020 to 2023 (Ex. Covid 19).

Robustness. These event study results are not only robust to excluding the Euro Crisis or post-Covid 19 observations. We obtain broadly similar results when considering various subsamples, distinguishing exchange rate regimes (Ilzetzi, Reinhart, and Rogoff, 2019), the degree of economic development (International Monetary Fund, 2023), and focusing only on Euro Area members. The corresponding results are provided in Figure B.3, Figure B.4, and Figure B.5, respectively. This underscores the robustness and broad applicability of our approach.

Predictability. To address residual concerns about potential pre-trends and the predictability of our surprises, we conduct a series of Granger causality tests. Throughout all Granger causality tests, we present standard errors in parentheses clustered at the country and time level. First, at daily frequency, we project each yield surprise on five daily lags of the bond yield and of stock prices, and report the results in Table 2. Across all yield surprise series, we find no statistically significant predictor at the 5% level, confirming the event study evidence on the absence of pre-trends.

Table 2: Predictability of daily yield change surprises

	<i>Dependent variable: Yield-change surprise</i>				
	$YCS_{i,t}$	$YCS_{i,t}^{\mathcal{D}}$	$YCS_{i,t}^{\mathcal{U}}$	$YCS_{i,t}^{\mathcal{H}}$	$YCS_{i,t}^{\mathcal{L}}$
Stock price ($t - 1$)	-0.690 (0.416)	-0.297 (0.479)	-0.793 (0.522)	-0.227 (0.344)	-0.417 (0.276)
Stock price ($t - 2$)	0.428 (0.595)	0.243 (0.788)	0.536 (0.742)	-0.117 (0.525)	0.325 (0.544)
Stock price ($t - 3$)	0.260 (0.400)	-0.427 (0.626)	0.686 (0.479)	0.093 (0.453)	0.180 (0.364)
Stock price ($t - 4$)	0.227 (0.356)	0.656 (0.559)	-0.034 (0.455)	0.138 (0.236)	0.352 (0.399)
Stock price ($t - 5$)	-0.225 (0.311)	-0.167 (0.401)	-0.396 (0.330)	0.102 (0.213)	-0.426 (0.269)
Bond yield ($t - 1$)	0.006 (0.056)	-0.068 (0.099)	0.128* (0.075)	0.079 (0.051)	-0.066 (0.069)
Bond yield ($t - 2$)	-0.009 (0.082)	0.049 (0.115)	-0.140 (0.125)	-0.035 (0.076)	0.081 (0.087)
Bond yield ($t - 3$)	-0.009 (0.072)	-0.016 (0.083)	0.043 (0.100)	-0.064 (0.052)	-0.042 (0.056)
Bond yield ($t - 4$)	0.020 (0.073)	0.048 (0.074)	-0.069 (0.168)	0.122 (0.075)	-0.002 (0.073)
Bond yield ($t - 5$)	-0.006 (0.063)	-0.011 (0.053)	0.034 (0.149)	-0.097* (0.053)	0.023 (0.043)
Adj. R^2	-0.008	-0.005	-0.008	0.130	0.192
Country FE	✓	✓	✓	✓	✓
Observations	1174	464	708	570	599

Notes: This table presents daily OLS regression results, as described in the text. The left-hand side variables are all non-zero yield change surprises. The regressors include a country fixed effect and daily lags of the 10-year sovereign bond yield and the main stock market index of the country under consideration. Column (1) shows all surprises pooled, columns (2) and (3) distinguish between surprises arising from credit rating downgrades ($YCS_{i,\tau}^{\mathcal{D}}$) vs. credit rating upgrades ($YCS_{i,\tau}^{\mathcal{U}}$), and columns (4) and (5) distinguish between positive ($YCS_{i,\tau}^{\mathcal{H}}$) vs. negative surprises ($YCS_{i,\tau}^{\mathcal{L}}$). Standard errors are in parentheses and clustered by country and time, and *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

We repeat this exercise but include ten daily lags of the same set of predictors and report the results in [Table B.2](#). Of the 100 coefficients we estimate, only three are statistically significant at the 5% level, which is well below the number of spuriously significant coefficients one would expect under the null of no predictability (Type I error rate).

Lastly, we use the aggregated monthly surprise variables and project them on lagged macroeconomic indicators to test whether our surprises are predictable at this lower frequency. We consider three monthly lags and a rich set of predictors, namely monthly bond yields, industrial production, unemployment, consumer prices, and producer prices, as well as lags of the

monthly surprise under consideration.¹⁷ The corresponding results are shown in [Table B.3](#). As before, we find no clear evidence of predictability. The adjusted R^2 is always below 0.1, and of the 90 estimated coefficients, only 3 are statistically significant at the 5% level, again well below the Type I error rate one would expect.

Overall, we find no clear evidence of predictability of our surprise series. Nevertheless, when estimating the macroeconomic effects of sovereign credit risk in [Section 5](#), we always include a set of controls to address potentially remaining concerns, and to enhance the signal-to-noise ratio.

Explanatory power. How well do our yield change surprises explain aggregate developments in sovereign bond yields? The event studies in [Figure 3](#) already suggested that our yield change surprises have a persistent effect on sovereign bond yields over the 10-day window under consideration. We now expand this analysis and estimate a stylized first stage, projecting monthly changes in 10-year sovereign bond yields on our surprises. Formally, we estimate

$$\Delta Y_{i,t} = \alpha_i + \beta YCS_{i,t}^k + \gamma X_{i,t} + v_{i,t}, \quad (7)$$

where $\Delta Y_{i,t}$ is the month-on-month change in the 10-year sovereign bond yield, α_i is a country fixed effect, $YCS_{i,t}^k$ is our surprise of interest aggregated to monthly frequency, $X_{i,t}$ is a vector of controls including three monthly lags of the left-hand side variable and of the surprise under consideration, and $v_{i,t}$ is an error term.

The regression results are provided in [Table 3](#), with standard errors in parentheses clustered at the country and time level. The surprise series delivers coefficients with the expected positive sign, ranging between 0.55 and 1.00. These estimates are all statistically significant at the 1% level, regardless of the surprise series under consideration. The partial R^2 , after

¹⁷Note that these data are only available for a subset of our sample; see [Table A.1](#).

Table 3: Yield change surprises and monthly sovereign bond yields

	Dependent variable: $\Delta \text{Bond Yield}_{i,t}$									
	All		Downgrades		Upgrades		Higher Yield		Lower Yield	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$YCS_{i,t}$	0.788*** (0.126)	0.781*** (0.127)								
$YCS_{i,t}^D$			0.798*** (0.144)	0.797*** (0.149)						
$YCS_{i,t}^U$					0.782*** (0.174)	0.767*** (0.173)				
$YCS_{i,t}^H$							1.002*** (0.125)	0.955*** (0.130)		
$YCS_{i,t}^L$									0.552*** (0.192)	0.551*** (0.189)
Partial R^2	0.003	0.003	0.001	0.001	0.002	0.002	0.002	0.002	0.001	0.001
Robust F-stat	39.2	37.8	30.7	28.5	20.2	19.7	64.7	54.3	8.3	8.5
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Lagged controls		✓		✓		✓		✓		✓
Observations	17520	17367	17520	17367	17520	17367	17520	17367	17520	17367

Notes: This table presents monthly OLS regression results based on Equation 7, as described in the text. The left-hand side variable is the monthly change in the 10-year sovereign bond yield in all columns. The regressors include the surprise of interest, a country fixed effect, and, when indicated, a vector of controls. The control vector comprises three monthly lags of the left-hand side and the surprise. Columns (1) and (2) use all surprises, columns (3) to (6) distinguish between surprises arising from credit rating downgrades ($YCS_{i,\tau}^D$) vs. credit rating upgrades ($YCS_{i,\tau}^U$), and columns (7) to (10) distinguish between positive ($YCS_{i,\tau}^H$) vs. negative surprises ($YCS_{i,\tau}^L$). Standard errors are in parentheses and clustered by country and time, and *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively. Since these estimates can be interpreted as a stylized first stage, we also present the (robust) Kleibergen-Paap F statistics.

partialling out controls and fixed effects, is comparatively low. This is to be expected given the large number of observations relative to only 1262 sovereign credit rating announcements.¹⁸ Yet, the Kleibergen-Paap F-statistic is relatively high for most surprise series, often well above 20. Only negative surprises (i.e., lower-yield surprises) display a relatively low F-statistic of around 8.¹⁹

Overall, these results provide clear evidence that yield change surprises significantly move sovereign bond yields, suggesting that they may have economically meaningful effects on financial markets and macroeconomic outcomes. The remainder of the paper investigates these effects.

¹⁸The results discussed above are based on the full sample. Restricting the estimation to non-zero surprises only, we obtain considerably higher R^2 values of up to 0.10. We also obtain equally significant results across all surprise series except for lower-yield surprises; see Table B.4.

¹⁹Note that in a just identified setting, the Kleibergen-Paap F-statistic is equivalent to the robust F-statistic from Montiel-Olea and Pflueger (2013), see also Andrews, Stock, and Sun (2019).

4 Sovereign credit risk and asset prices

In this section, we investigate the relationship between shocks to sovereign credit risk, as measured by our yield change surprises, and asset prices in financial markets. We find that increases in sovereign credit risk are associated with significant and simultaneous declines in stock market valuations and corporate bond prices.

4.1 Econometric specification

Sovereign credit risk may propagate to the broader economy partly via its effect on asset prices and financial conditions. We assess this channel through high-frequency event studies, as pioneered by [Cook and Hahn \(1989\)](#) and widely applied to monetary policy (e.g., [Kuttner, 2001](#); [Bauer and Swanson, 2023b](#)) and macroeconomic data releases (e.g., [Gürkaynak et al., 2020](#); [Hack and Rostam-Afschar, 2025](#)). The key idea is to estimate the effect of the high-frequency yield surprise on changes in asset prices, measured over the same two-day window as the yield change on which the surprise is based. Formally, we estimate

$$y_{i,\tilde{\tau}+1} - y_{i,\tilde{\tau}-1} = \alpha_i + \alpha_{q(\tilde{\tau})} + \beta YCS_{i,\tilde{\tau}} + \gamma X_{i,\tilde{\tau}} + \varepsilon_{i,\tilde{\tau}}, \quad (8)$$

where the subscript $\tilde{\tau}$ indexes only trading days with a non-zero yield change surprise, i.e., $YCS_{i,\tilde{\tau}} \neq 0$. We include country fixed effects α_i and quarterly time fixed effects $\alpha_{q(\tilde{\tau})}$, as well as a vector of controls comprising five daily lags of the outcome variable $y_{i,\tilde{\tau}}$ and five daily lags of the 10-year sovereign bond yield. Lastly, $\varepsilon_{i,\tilde{\tau}}$ is an error term.

Matching the surprise as constructed in [Section 3](#), the left-hand side variable is the two-day change in either a country’s main stock market index or a proxy for corporate bond prices, capturing corporate funding costs.²⁰ To ease interpretation and improve comparability, we standardize the left-hand side variable so that it has unit variance within each country.

²⁰This proxy is either an index of high-grade corporate bond prices or an ETF comprising such securities.

Additionally, we winsorize the left-hand side variable at the $\pm 1\%$ level to mitigate the effect of outliers.²¹

This specification is appealing because it differences out potential pre-trends on both the left-hand side and the right-hand side, by virtue of using a surprise variable and estimating its effect within an equally narrow window. As a result, the estimates robustly capture the revision in asset prices associated with a change in sovereign credit risk, as measured by the yield change surprise.

4.2 Stock markets

Based on this specification, we first estimate the response of stock markets to sovereign credit risk surprises and report the results in [Table 4](#), with standard errors clustered at the country and time level in parentheses. Columns (1), (4), and (7) present a parsimonious specification with country fixed effects only. In the remaining columns, we incrementally add time fixed effects and lagged controls, as described in [Section 4.1](#).

Pooled evidence. Columns (1) to (3) present the pooled evidence using all yield change surprises, regardless of their sign. Across all specifications, we find a statistically significant decline in stock markets at the 1% level. These estimates are remarkably stable across specifications. Quantitatively, a one percentage point increase in sovereign bond yields due to a reassessment of sovereign credit risk is associated with a one standard deviation decline in stock market valuations. This effect can be considered modest, since a one percentage point surprise corresponds to approximately 10 standard deviations of the yield change surprises; see [Table 1](#). We revisit this result when considering the effects of sovereign credit risk at business cycle frequency in [Section 5](#).

²¹Formally, we set values below (above) the 1 percentile (99 percentile) to the value of that percentile. This is a minimal increase from the $\pm 0.5\%$ winsorization used in [Figure 2](#). We do so because the samples in this section tend to be small, giving potentially outlying observations greater weight, and because winsorizing at the 0.5% level has no effect in samples smaller than $N = 200$, since the most extreme observation and the threshold would coincide. For maximum transparency, we also report all results in their unaltered form in [Appendix C](#).

Table 4: Stock market responses to sovereign credit risk

	<i>Dependent variable: Stock Price Surprise</i>								
	All			Higher-Yield			Lower-Yield		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$YCS_{i,t}$	-1.084*** (0.264)	-0.936*** (0.254)	-0.959*** (0.252)						
$YCS_{i,t}^H$				-1.514*** (0.435)	-1.599*** (0.504)	-1.457*** (0.458)			
$YCS_{i,t}^L$							-0.691 (0.472)	-1.847*** (0.542)	-1.832*** (0.586)
Within R^2	0.015	0.011	0.011	0.017	0.019	0.015	0.003	0.023	0.021
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE		✓	✓		✓	✓		✓	✓
Lagged Controls			✓			✓			✓
Observations	1164	1154	1153	569	557	557	595	579	578

Notes: This table presents event study OLS estimates based on Equation 8 at trading day frequency, as specified in Section 4.1. The left-hand side variable is the change in the stock price index over a two-day window, matching the event window over which the yield change surprises are computed. The regressors include the surprise of interest, a country fixed effect, and, when indicated, quarterly time fixed effects and a vector of controls. The control vector comprises five daily lags of the asset price underlying the left-hand side and five daily lags of the 10-year sovereign bond yield. Columns (1) to (3) use all surprises, columns (4) to (6) consider positive surprises ($YCS_{i,\tau}^H$), and columns (7) to (9) consider negative surprises ($YCS_{i,\tau}^L$). Standard errors are in parentheses and clustered by country and time, and *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

Surprise heterogeneity. In columns (4) to (9), we use only rating changes with positive surprises (columns (4) to (6)) or negative surprises (columns (7) to (9)) to assess potential sign-dependence. In the parsimonious specification, we find that increases in sovereign credit risk (i.e., higher yield surprises) lead to a stronger decline in stock markets compared to decreases in sovereign credit risk; see columns (4) and (7). However, when including quarterly time fixed effects, this asymmetry vanishes, though the effects on stock markets remain stronger than in the pooled specification. The reason for this difference lies in the nature of our surprise series. Sovereign credit rating announcements are relatively rare, so some quarters contain only a single sovereign credit rating announcement across all countries. These isolated events are absorbed by the quarterly time fixed effects. This points toward a further non-linearity: sovereign credit risk may have more detrimental effects when multiple countries are affected simultaneously, as was, for example, the case during the Euro Crisis. When splitting the sample by sign, we increase the proportion of these types of events, as singletons become more likely in the smaller samples and are absorbed. We may therefore conclude that the pooled surprise delivers a somewhat conservative estimate of the consequences of sovereign credit risk, as it averages across heterogeneous episodes.

Cross-country heterogeneity. Another potentially important dimension of heterogeneity pertains to more slowly moving country characteristics. To this end, we investigate heterogeneity between emerging and developed economies as well as between countries with floating and fixed exchange rate regimes. These results are presented in [Table C.5](#) and [Table C.6](#) in [Appendix C](#), respectively. Here, the focus of our discussion and interpretation rests on the pooled specifications. We also report results split by the sign of the surprise, though statistical power is limited in these cases, as the sample becomes small when studying sign-dependence and cross-country heterogeneity simultaneously.

Focusing on the degree of economic development, we find coefficients of comparable magnitude for developed and emerging economies, with emerging economies displaying slightly more statistically significant estimates. For both groups, a one percentage point increase in sovereign bond yields is associated with approximately one standard deviation reduction in stock prices, consistent with the baseline result. As sample sizes become small when splitting simultaneously by the sign of the surprise and the type of economy, we are cautious about overemphasizing these findings, but note that coefficients are negative across all specifications and are particularly significant for developed economies.

When discriminating between exchange rate regimes, we again find similar magnitudes and high statistical significance for the pooled coefficients across both fixed and floating rate regimes, with coefficients of slightly larger magnitude than in the baseline, ranging from 1.2 to 1.6 standard deviations. When additionally splitting by the sign of the surprise, we again find negative coefficients across all specifications, though statistical power varies more across specifications.

Overall, a negative stock market response to an increase in sovereign credit risk appears to be a robust feature of the data. This finding holds across the sign of the surprise and across several dimensions of cross-country heterogeneity, underscoring that sovereign credit risk may have important adverse macroeconomic consequences.

4.3 Corporate funding costs

So far, we have studied stock prices, which are informative about the anticipated macroeconomic effects of sovereign credit risk. Next, we investigate one potential channel through which sovereign credit risk may transmit to the real economy more directly. An increase in sovereign credit risk may partly propagate by worsening credit conditions for the private sector. We therefore now study the response of corporate funding costs, as captured by measures of corporate bond prices.²² Specifically, we estimate the response of corporate bond prices based on [Equation 8](#) and report the corresponding results in [Table 5](#).

We find that corporate bond prices decline substantially, with the response being considerably more pronounced than for stock prices. A one standard deviation increase in sovereign yields (10 basis points) is associated with a decline in corporate bond prices between 0.4 and 0.6 standard deviations (recall that the coefficients are scaled to correspond to a 1 percentage point yield increase). As reliable proxies for corporate bond prices are only available at the required daily frequency for large advanced economies, the sample is smaller and reflects this selected subgroup rather than the broad mix of advanced and emerging economies in the baseline stock market specification.

This finding of larger effects for corporate bond prices is consistent with the concept of a sovereign yield floor (sovereign risk ceiling), whereby the sovereign borrows at the lowest rate within the domestic economy, setting a lower bound on private sector borrowing costs ([Durbin and Ng, 2005](#); [Almeida et al., 2017](#)). When sovereign yields rise, corporate borrowing costs increase accordingly, pushing down corporate bond prices. Conversely, when yields fall, corporate bond prices rise, as reflected in the sign-dependent coefficients in columns (4) to (9). This channel is also reflected in the substantially higher *within* R^2 of up to 0.36, compared with the low R^2 when studying stock market responses. This increase in funding cost may translate directly into a negative supply shock to the economy, as higher borrowing costs

²²These measures are heterogeneous across countries and dependent on data availability. As discussed above, we use either an index of high-grade corporate bond prices or ETFs tracking such securities. To obtain a comparable measure across countries, these variables are standardized at the country level.

raise operating costs for corporates. We revisit this possibility in the next section when we estimate the macroeconomic effects of sovereign credit risk.

Table 5: Corporate bond price responses to sovereign credit risk

	<i>Dependent variable: Corporate Bond Price Surprise</i>								
	All			Higher-Yield			Lower-Yield		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$YCS_{i,t}$	-5.920*** (1.193)	-6.483*** (1.154)	-5.980*** (0.922)						
$YCS_{i,t}^H$				-5.747*** (1.387)	-5.980*** (1.445)	-3.587*** (0.939)			
$YCS_{i,t}^L$							-6.478*** (1.259)	-4.249** (1.577)	-1.945 (1.540)
Within R^2	0.336	0.361	0.353	0.243	0.297	0.158	0.230	0.109	0.031
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE		✓	✓		✓	✓		✓	✓
Lagged Controls									✓
Observations	232	218	218	117	90	90	115	85	85

Notes: This table presents event study OLS estimates based on Equation 8 at trading day frequency, as specified in Section 4.1. The left-hand side variable is the change in corporate bond prices over a two-day window, matching the event window over which the yield change surprises are computed. The regressors include the surprise of interest, a country fixed effect, and, when indicated, quarterly time fixed effects and a vector of controls. The control vector comprises five daily lags of the asset price underlying the left-hand side and five daily lags of the 10-year sovereign bond yield. Columns (1) to (3) use all surprises, columns (4) to (6) consider positive surprises ($YCS_{i,\tau}^H$), and columns (7) to (9) consider negative surprises ($YCS_{i,\tau}^L$). Standard errors are in parentheses and clustered by country and time, and *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

5 The macroeconomic effects of sovereign credit risk

We estimate the macroeconomic effects of sovereign credit risk and find that changes in sovereign credit risk act as negative supply shocks: real activity falls while consumer prices rise in response to an increase in sovereign credit risk, as measured by our surprises.

5.1 Econometric specification

We use monthly local projections following Jordà (2005) to estimate the dynamic causal effect of sovereign credit risk, as captured by our surprises. To do so, we model our yield change surprises as a shock and use OLS to estimate the following sequence of regressions

$$y_{i,t+h} - y_{i,t-1} = \alpha_i^h + \alpha_{q(t)}^h + \beta^h YCS_{i,t} + \gamma^h X_{i,t} + \epsilon_{i,t}^h, \quad (9)$$

for $h = 0, \dots, 36$. α_i^h and $\alpha_{q(t)}^h$ denote country and quarterly time fixed effects, $YCS_{i,t}$ is the yield change surprise, $X_{i,t}$ is a vector of controls including three lags of the dependent variable $y_{i,t}$ as well as three lags of the 10-year sovereign bond yield. Finally, $\epsilon_{i,t}^h$ is an error term.

As left-hand side variables, we consider the monthly change between $t + h$ and $t - 1$ for the 10-year sovereign bond yield, the stock price index, the consumer price index, an industrial production index, and the number of unemployed individuals. All of these variables are expressed in natural logarithms (except for the bond yield), so that all estimates have a percent-change interpretation (percentage points for the bond yield). The estimates are further scaled to correspond to a 0.2 percentage point increase in sovereign bond yields on impact, which roughly corresponds to a two standard deviation surprise.²³

5.2 Macroeconomic effects

We present the estimation results for prices and real activity in [Figure 4](#) and [Figure 5](#), respectively. The solid blue lines indicate point estimates and the blue-shaded areas are the corresponding 95% confidence bands, based on standard errors clustered at the country and time level.

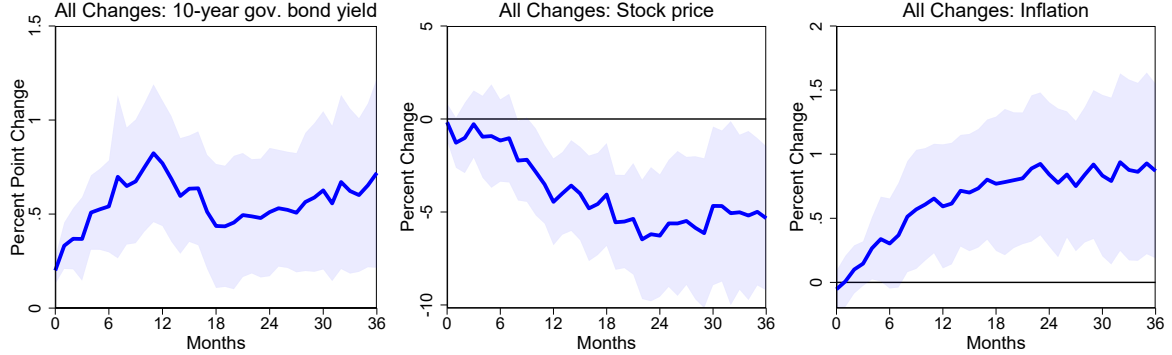
Responses of prices. Focusing on the responses of prices (or bond yields), we find statistically significant responses for all variables under consideration, as displayed in [Figure 4](#).

Sovereign bond yields increase from the initial level of 0.20 percentage points to around 0.75 percentage points after one year and slightly revert after this peak. However, over the three-year horizon under consideration, sovereign bond yields remain persistently elevated at around 0.5 percentage points. This suggests that sovereign credit risk has very persistent effects on *nominal* yields.

Stock prices do not respond significantly on impact, consistent with the modest event study

²³Note that expressing all estimates relative to the impact effect of bond yields also removes potential attenuation bias due to possible measurement error in our surprises.

Figure 4: Responses of prices to sovereign credit risk shocks



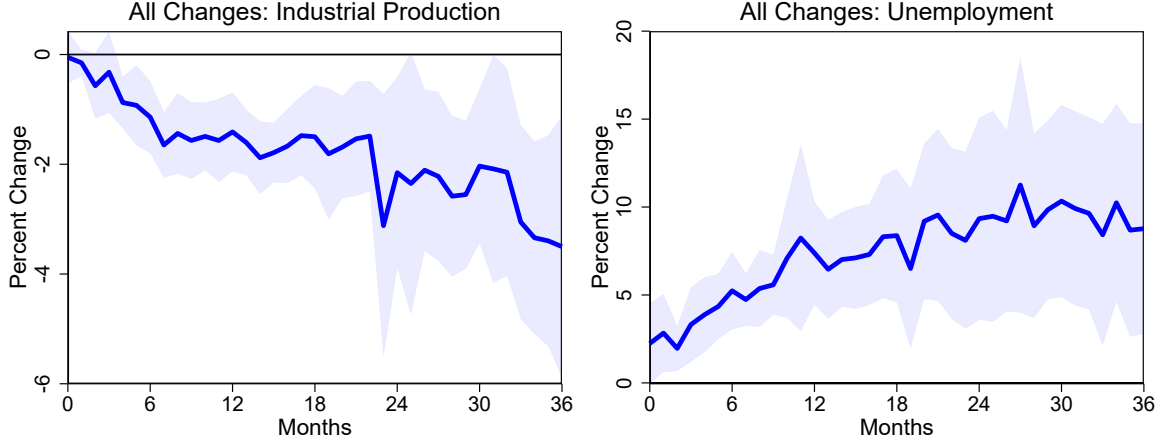
Notes: The figure displays monthly OLS local projection estimates based on Equation 9, as specified in Section 5.1. The left-hand-side variables are 10-year sovereign bond yields, stock price indices, and consumer prices and the regressor of interest is the yield change surprise. All estimates correspond to an initial yield increase of 0.2 percentage points. We further include country and time fixed effects, three lags of the left-hand side variable and the bond yield as controls. The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels.

effects documented in Section 4. However, after about 6 months, stock market valuations fall consistently and plateau at a lower level after two years, with stock market indices about 5 percent lower in nominal terms.

The right panel of Figure 4 shows the response of consumer prices, which increase steadily. After three years, consumer prices are about 1 percent higher. This price response reveals that the real stock market decline is larger than the nominal stock price response suggests. For sovereign bond yields, the price response implies that ex-post *real* yields increase by less than nominal yields.

There are three potentially relevant channels through which sovereign credit risk raises consumer prices. First, as documented in Section 4.3, higher private funding costs may induce firms to raise prices. Second, through the lens of fiscal dominance (e.g., Leeper, 1991; Sims, 1994; Cochrane, 2001), private investors may discount future primary surpluses of the government more strongly, so that an increase in the price level may be needed to balance the government's intertemporal budget constraint in equilibrium. Third, the persistent *nominal* yield response points to a Fisher channel, in which permanent or highly persistent increases in interest rates induce higher inflation when equilibrium conditions require a lower *real* rate (e.g., Mishkin, 1992; Uribe, 2022).

Figure 5: Responses of real activity to sovereign credit risk shocks



Notes: The figure displays monthly OLS local projection estimates based on Equation 9, as specified in Section 5.1. The left-hand-side variables are industrial production and unemployment and the regressor of interest is the yield change surprise. All estimates correspond to an initial yield increase of 0.2 percentage points. We further include country and time fixed effects, three lags of the left-hand side variable and the bond yield as controls. The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels.

Responses of real activity. Given the persistent effects on sovereign bond yields, stock market valuations, and consumer prices, we next investigate whether sovereign credit risk has real adverse consequences. To this end, we study the responses of industrial production and unemployment in Figure 5. We find that sovereign credit risk indeed has a strong and statistically significant effect on both variables. Industrial production steadily declines and is almost 3 percent lower after three years. Similarly, unemployment increases by around 10 percent. This magnitude appears somewhat large. Importantly, however, note that this is a percent change and not a percentage point change.²⁴

Overall, the estimates confirm that sovereign credit risk has real adverse consequences. These negative effects may operate partly via aggregate demand, for example, through reduced government spending and private investment due to higher funding costs, and possibly also due to elevated uncertainty. However, as shown above, the response of consumer prices is dominated by channels other than aggregate demand. As a result, sovereign credit risk acts as a negative supply shock, raising consumer prices while lowering economic activity.

²⁴If we assume the much larger number of employed individuals to remain relatively stable in relation to the number of unemployed, the interpretation would be, for example, an increase from 5 to 5.5 percent.

Surprise heterogeneity. We further investigate whether there is significant heterogeneity depending on the sign of the surprise, and report the results in [Figure D.1](#) and [Figure D.2](#). Focusing on prices, we find that increases in sovereign credit risk lead to persistent increases in sovereign bond yields, a corresponding decline in stock prices, and an increase in consumer prices. These results are similar to the baseline but somewhat noisier. Yet we still obtain statistically significant effects at the 5% level for many response horizons. In contrast, when focusing on negative surprises that lower sovereign bond yields, we find largely insignificant effects on sovereign bond yields and stock prices, although we still obtain significant and strong effects on consumer prices.

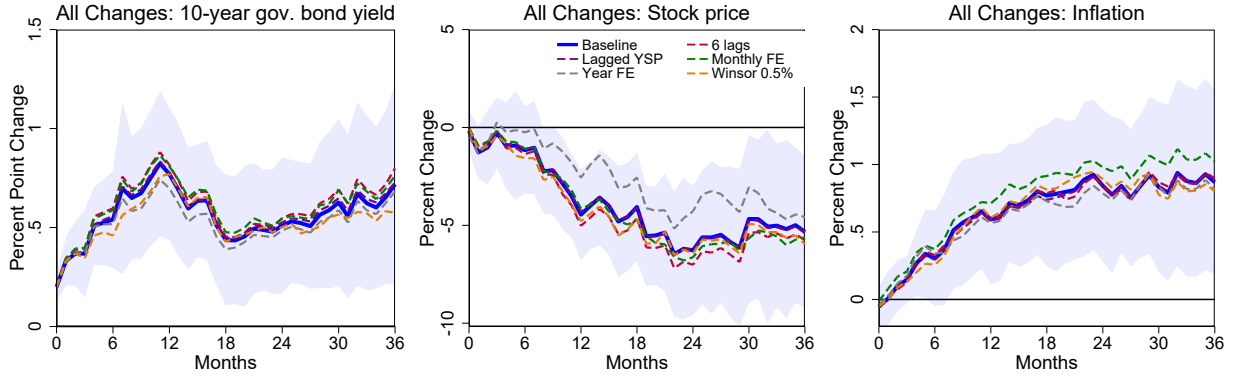
When studying real activity responses to higher-yield surprises, we obtain estimates that are comparable to the pooled baseline in terms of shape, magnitude, and statistical significance. Conversely, for lower-yield surprises, capturing reductions in sovereign credit risk, we obtain no significant responses of industrial production and unemployment, though point estimates trend in the same direction as in the baseline.

Overall, this analysis shows that our baseline results are particularly driven by unexpected increases in sovereign credit risk, whereas we do not have sufficient statistical power to obtain meaningful estimates for the consequences of a sovereign credit risk reduction. Note that we do not take a definitive stance on why lower-yield surprises yield inconclusive results. This could be because more credible government policies take longer to materialize in terms of macroeconomic outcomes, but also because changes in sovereign credit ratings may provide less powerful identifying variation for these types of events. To this end, recall that lower-yield surprises had the lowest F-statistic in the stylized first stage; see [Table 3](#).

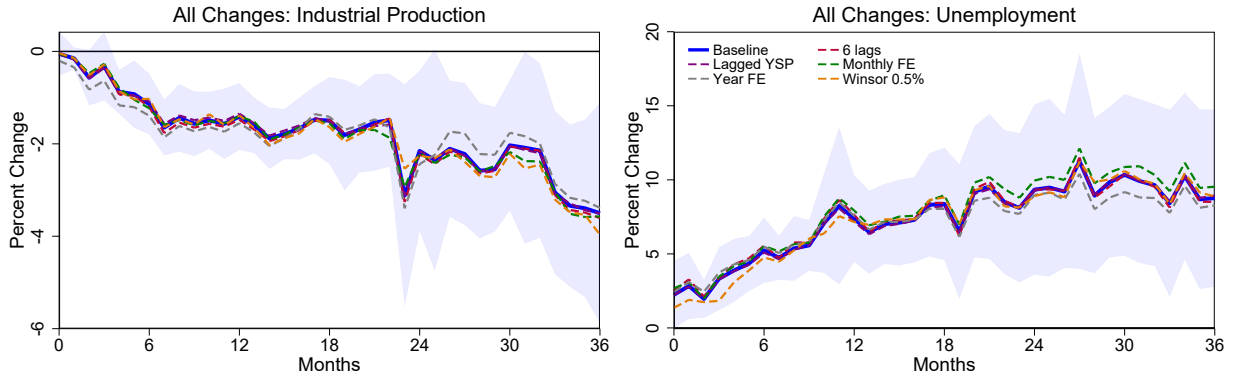
Robustness. As a final step of our analysis, we establish that our main results for macroeconomic outcomes in [Figure 4](#) and [Figure 5](#) are remarkably robust across various specification and modelling choices. To this end, we reproduce the baseline responses for our five main outcomes (in blue) in [Figure 6](#). Panel A shows the responses of prices (bond yields, stock

Figure 6: Responses of prices and real activity to sovereign credit risk shocks, robustness

(a) Responses of prices to sovereign credit risk shocks



(b) Responses of real activity to sovereign credit risk shocks



Notes: The figure displays monthly OLS local projection estimates based on Equation 9, as specified in Section 5.1. The left-hand-side variables are 10-year sovereign bond yields, stock price indices, consumer prices, industrial production and unemployment and the regressor of interest is the yield change surprise. All estimates correspond to an initial yield increase of 0.2 percentage points. The blue lines and shaded areas reproduce the baseline specification, which includes country and quarterly time fixed effects and three lags of the left-hand side variable and the bond yield as controls. The remaining lines show alternative specifications: extending the baseline controls from three to six lags (red); adding three lags of the yield change surprise to the baseline controls (purple); replacing quarterly time fixed effects with monthly fixed effects (green); replacing quarterly time fixed effects with yearly fixed effects (grey); and winsorizing the left-hand side variable at 0.5% (orange). The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels.

prices, and consumer prices), and Panel B shows the responses of industrial production and unemployment. We then overlay the baseline specification with a set of alternative estimates. The first alternative specification, plotted in red, extends the baseline set of lagged controls from 3 to 6 lags, to account for potentially longer trends in bond yields and the outcome prior to a yield change surprise. The second specification, plotted in purple, adds the yield change surprise itself to the baseline set of lagged controls to account for autocorrelation in our surprise variable. As previously discussed, the distribution of our surprises, with a potentially low number of events within a given time window, may be sensitive to the choice

of time fixed effects. To alleviate the concern that this sensitivity unduly influences our results, we estimate specifications with monthly (in green) and yearly (in grey), rather than the quarterly fixed effects used in the baseline. Finally, to account for the possibility that a few extreme events drive our aggregate results, we winsorize the left-hand side variables at 0.5% and plot the results in orange.

Across all these specifications and across all five outcomes, the results remain stable and very close to the baseline, confirming that our findings are not driven by the choice of specification. When deviations from the baseline occur, they tend to be small and, as expected, arise for the alternative fixed effects specifications.

For completeness, [Figure D.3](#) and [Figure D.4](#) show the same set of robustness specifications for financial and real outcomes for sign-dependent yield change surprises.

As a final exercise, [Figure D.5](#) and [Figure D.6](#) present specifications in which we control for three lags of all presented outcomes, to account for potential interdependence between them. Note that this also fixes the sample to the set of observations for which all outcomes are available. Despite this large set of controls and the reduction in observations (particularly for bond yields and stock prices, for which we have the most extensive coverage) the baseline results are confirmed. Sovereign bond yields and the price level rise persistently, and stock prices decline. Similarly, industrial production declines while unemployment rises, and again the results can be attributed mostly to positive yield change surprises, i.e., unexpectedly higher yields.

6 Conclusion

This paper proposes a novel high-frequency identification approach to estimate the macroeconomic consequences of sovereign credit risk. Using changes in sovereign bond yields in tight windows around credit rating announcements by private rating agencies, we construct yield change surprises for a large cross-section of economies spanning several decades. These

surprises are plausibly exogenous to macroeconomic fundamentals, not predictable by past macroeconomic data, and have meaningful explanatory power for sovereign bond yields.

We use these surprises to study two sets of outcomes. In daily event studies, we find that increases in sovereign credit risk are associated with significant declines in stock market valuations and corporate bond prices, suggesting that financial markets anticipate contractionary consequences and that sovereign credit risk raises private funding costs. In monthly local projections, we find that sovereign credit risk acts as a negative supply shock: economic activity falls persistently while consumer prices rise. These effects are robust across a wide range of specification choices and are primarily driven by unexpected increases in sovereign credit risk. Taken together, the results suggest that sovereign credit risk has significant and persistent adverse consequences for both financial markets and the real economy.

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Appendix

The Macroeconomic Consequences of Sovereign Credit Risk

June 9, 2026

Lukas Diebold and Lukas Hack

Appendix A Data

Figure A.1: Credit ratings of the United States (May 2026)

Grade	Description	S&P	Moody's	Fitch	DBRS
Investment Grade	Prime	AAA	Aaa	AAA	AAA
	High Medium Grade	AA+	Aa1	AA+	AA(high)
		AA	Aa2	AA	AA
		AA-	Aa3	AA-	AA(low)
	Upper Medium Grade	A+	A1	A+	A(high)
		A	A2	A	A
		A-	A3	A-	A(low)
	Lower Medium Grade	BBB+	Baa1	BBB+	BBB(high)
		BBB	Baa2	BBB	BBB
		BBB-	Baa3	BBB-	BBB(low)
Speculative Grade	Speculative	BB+	Ba1	BB+	BB(high)
		BB	Ba2	BB	BB
		BB-	Ba3	BB-	BB(low)
	Highly Speculative	B+	B1	B+	B(high)
		B	B2	B	B
		B-	B3	B-	B(low)
	Substantial Risk	CCC+	Caa1	CCC+	CCC(high)
		CCC	Caa2	CCC	CCC
		CCC-	Caa3	CCC-	CCC(low)
	Extremely Speculative	CC	Ca	CC	CC
		C	C	C	C
	In Default	RD	/	RD	RD
		SD	/	SD	SD
		D•NR	D•NR	D•NR	D•NR

Notes: This figure shows the rating scale of the four leading rating agencies. The exhibit is taken from the website of [World Government Bonds Credit Ratings \(2026\)](#) and indicates the sovereign credit rating of the United States as of May 2026 with green circles.

Table A.1: Summary table

Country	Monthly			Daily				Intraday
	Ind. Produc- tion	Unemployment	CPI	Rating	Bond Yield	Stock Index	Corp. Bond	Bond Yield
Australia	-	1978-2023	-	1960-2026	1976-2025	1992-2026	2020-2026	2021-2026
Austria	1960-2023	1994-2023	1960-2025	1975-2026	1991-2025	1986-2026	2008-2026	2021-2026
Belgium	1960-2024	1983-2023	1960-2025	1988-2026	1993-2025	1991-2026	-	2021-2026
Brazil	-	-	-	1986-2026	1998-2025	1985-2026	-	2021-2026
Bulgaria	-	-	-	1996-2026	2002-2025	2000-2026	-	-
Canada	1961-2024	1976-2023	1960-2025	1960-2026	1994-2025	1981-2026	2006-2026	2021-2026
Chile	1991-2024	2005-2023	1970-2023	1992-2026	2007-2025	-	-	2021-2026
China	-	-	-	1988-2026	2000-2025	1990-2026	-	2021-2026
Colombia	-	-	-	1993-2026	2002-2025	2008-2026	-	2021-2026
Croatia	-	-	-	1997-2026	2008-2025	1998-2026	-	2025-2026
Cyprus	-	-	-	1994-2026	2015-2025	1993-2026	-	2021-2026
Czech Republic	1990-2024	1993-2023	1991-2025	1993-2026	2000-2025	2006-2026	-	2021-2026
Denmark	1974-2024	1983-2023	1967-2025	1967-2026	1994-2025	1989-2026	-	2021-2026
Estonia	-	2000-2023	-	-	-	1996-2026	-	-
Finland	1960-2024	1988-2023	1960-2025	1972-2026	1994-2025	1988-2026	1998-2026	2021-2026
France	1960-2024	1983-2023	1960-2025	1975-2026	1986-2025	1987-2026	1998-2026	2021-2026
Germany	1960-2024	1991-2023	1960-2025	1983-2026	1959-2025	1987-2026	1998-2026	2021-2026
Greece	1962-2024	1998-2023	1960-2025	1999-2026	1998-2025	1991-2026	-	2021-2026
Hong Kong	-	-	-	1988-2026	1998-2025	1970-2026	-	2021-2026
Hungary	1985-2024	1995-2023	1980-2025	1989-2026	2003-2025	1991-2026	-	2021-2026
Iceland	-	2003-2023	1960-2025	1989-2026	2002-2025	2004-2026	-	2021-2026
India	-	-	-	1988-2026	1998-2025	1995-2026	-	2021-2026
Indonesia	-	-	-	1992-2026	2003-2025	1990-2026	-	2021-2026
Ireland	1975-2023	1983-2023	1960-2025	1987-2026	1998-2025	1983-2026	-	2021-2026
Israel	1990-2023	2012-2023	1970-2025	1988-2026	2002-2025	1992-2026	-	2024-2026
Italy	1960-2024	1983-2023	1960-2025	1986-2026	1991-2025	1997-2026	2009-2026	2021-2026
Japan	1960-2024	1970-2023	1960-2021	1960-2026	1966-2025	1979-2026	2016-2026	2021-2026
Latvia	-	-	-	1997-2026	-	2000-2026	-	-
Lithuania	-	-	-	1996-2026	2006-2025	2000-2026	-	2021-2026
Malaysia	-	-	-	1986-2026	2001-2025	1982-2026	-	2021-2026
Malta	-	-	-	1994-2026	2008-2025	1995-2026	-	-
Mexico	-	2005-2023	1969-2024	1990-2026	2001-2025	1987-2026	-	2021-2026
Netherlands	1960-2023	1983-2023	1960-2025	1986-2026	1993-2025	1983-2026	2008-2026	2021-2026
New Zealand	-	-	-	1961-2026	1994-2025	2000-2026	-	2021-2026
Norway	1960-2023	1989-2023	1960-2025	1960-2026	1994-2025	1999-2026	2008-2026	2021-2026
Peru	-	-	-	1996-2026	2010-2025	-	-	2021-2026
Philippines	-	-	-	1993-2026	2000-2025	1987-2026	-	2021-2026
Poland	1985-2024	1995-2023	1989-2025	1995-2026	1999-2025	1994-2026	-	2021-2026
Portugal	1960-2024	1983-2023	1960-2025	1986-2026	1994-2025	1992-2026	2008-2026	2021-2026
Romania	-	-	-	1996-2026	2007-2025	-	-	2021-2026
Russia	-	-	-	1996-2026	-	1997-2026	-	-
Singapore	-	-	-	1989-2026	1998-2025	1999-2026	-	2021-2026
Slovakia	1989-2024	1995-2023	1991-2025	1994-2026	2007-2025	1979-2026	-	2021-2026
Slovenia	1992-2024	1995-2023	1980-2025	1996-2026	2007-2025	2006-2026	-	2025-2026
South Africa	-	-	-	1994-2026	1995-2025	1995-2026	-	2024-2026
South Korea	1990-2024	1999-2023	1960-2023	1986-2026	2000-2025	1981-2026	2022-2026	2021-2026
Spain	1965-2023	1983-2023	1960-2025	1988-2026	1994-2025	1991-2026	1998-2026	2021-2026
Sweden	1960-2023	1983-2023	1960-2025	1977-2026	1997-2025	1986-2026	2008-2026	2021-2026
Switzerland	-	-	1960-2025	1982-2026	1994-2025	1988-2026	2014-2026	2021-2026
Taiwan	-	-	-	1989-2026	1998-2025	1969-2026	-	2021-2026
Thailand	-	-	-	1989-2026	2001-2025	1982-2026	-	2021-2026
Turkey	1985-2024	2005-2023	1960-2025	1993-2026	2011-2025	1988-2026	-	2021-2026
United Kingdom	1960-2024	1983-2023	1960-2025	1978-2026	1979-2025	1984-2026	1999-2026	2021-2026
United States	1960-2024	1960-2023	1960-2025	1960-2026	1959-2025	1969-2026	1999-2026	2021-2026

Notes: The table presents the sample coverage by country for the main variables of our analysis.

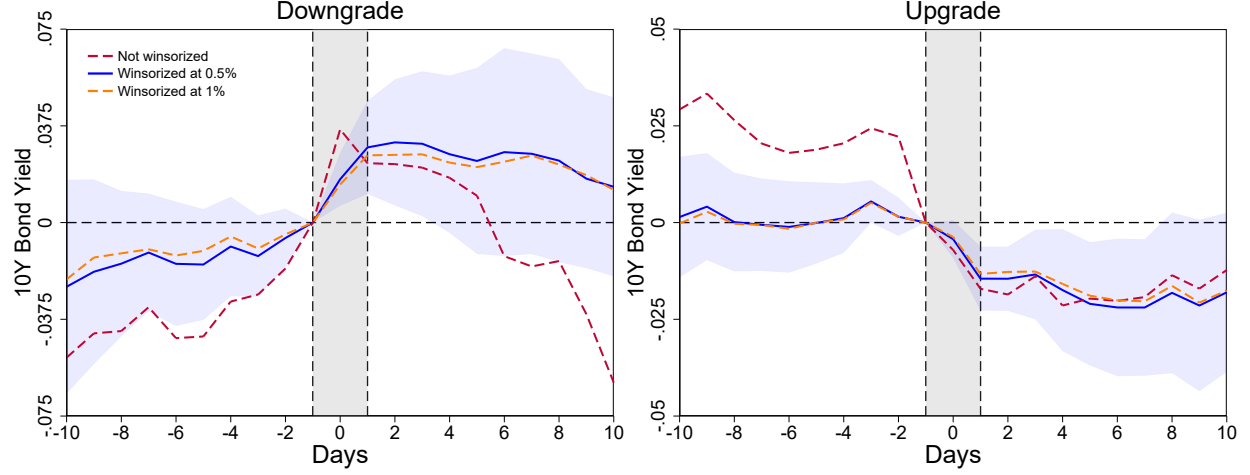
Table A.2: Summary statistics

	Mean	Median	Std. Dev.	Min	Max	Obs	Countries
Monthly							
Rating Index _{<i>i,t</i>}	25.134	26.250	6.030	1.333	31.500	23,047	53
10-Y Gov. Yield _{<i>i,t</i>}	5.302	4.605	3.600	-0.975	32.440	19,568	51
ln(Stock Price) _{<i>i,t</i>}	7.741	7.786	1.737	-11.513	11.995	20,164	50
ln(Corp. Bond Price) _{<i>i,t</i>}	5.019	4.984	0.921	2.909	11.676	3,871	17
ln(Ind. Prod.) _{<i>i,t</i>}	4.411	4.559	0.391	2.372	5.425	11,980	27
ln(Unemp.) _{<i>i,t</i>}	13.061	12.927	1.420	6.908	16.702	11,772	30
ln(CPI) _{<i>i,t</i>}	4.317	4.450	0.557	-1.538	7.063	13,637	30
Δ Rating Index _{<i>i,t</i>}	0.002	0.000	0.243	-15.667	12.000	23,023	53
Δ 10-Y Gov. Yield _{<i>i,t</i>}	-0.010	-0.010	0.398	-10.180	8.955	19,525	51
Δ ln(Stock Price) _{<i>i,t</i>}	0.007	0.008	0.075	-1.644	1.743	20,130	50
Δ ln(Corp. Bond Price) _{<i>i,t</i>}	0.002	0.002	0.014	-0.092	0.316	3,854	17
Δ ln(Ind. Prod.) _{<i>i,t</i>}	0.002	0.002	0.030	-0.400	0.401	11,977	27
Δ ln(Unemp.) _{<i>i,t</i>}	0.000	0.000	0.070	-1.253	1.253	11,754	30
Δ ln(CPI) _{<i>i,t</i>}	0.003	0.002	0.008	-0.031	0.210	13,635	30
Daily							
Rating Index _{<i>i,t</i>}	24.889	25.750	6.189	0.000	31.500	473,191	53
10-Y Gov. Yield _{<i>i,t</i>}	4.918	4.312	3.446	-1.107	39.850	375,193	51
ln(Stock Price) _{<i>i,t</i>}	7.739	7.784	1.734	-12.001	12.010	438,147	50
ln(Corp. Bond Price) _{<i>i,t</i>}	5.016	4.983	0.913	2.859	11.678	83,762	17
Δ Rating Index _{<i>i,t</i>}	0.000	0.000	0.041	-6.167	12.000	473,173	53
Δ 10-Y Gov. Yield _{<i>i,t</i>}	-0.000	0.000	0.115	-19.914	9.156	375,145	51
Δ ln(Stock Price) _{<i>i,t</i>}	0.000	0.000	0.021	-1.685	1.681	438,113	50
Δ ln(Corp. Bond Price) _{<i>i,t</i>}	0.000	0.000	0.003	-0.064	0.317	83,745	17
5-Minute Intraday							
10-Y Gov. Yield _{<i>i,t</i>}	4.241	3.330	3.892	-1.980	90.710	18,354,076	49
Δ 10-Y Gov. Yield _{<i>i,t</i>}	0.000	0.000	0.036	-88.365	88.371	18,341,325	49

Notes: The table presents summary statistics for the main variables of our analysis. All variables are included twice, once as (log) levels and changes.

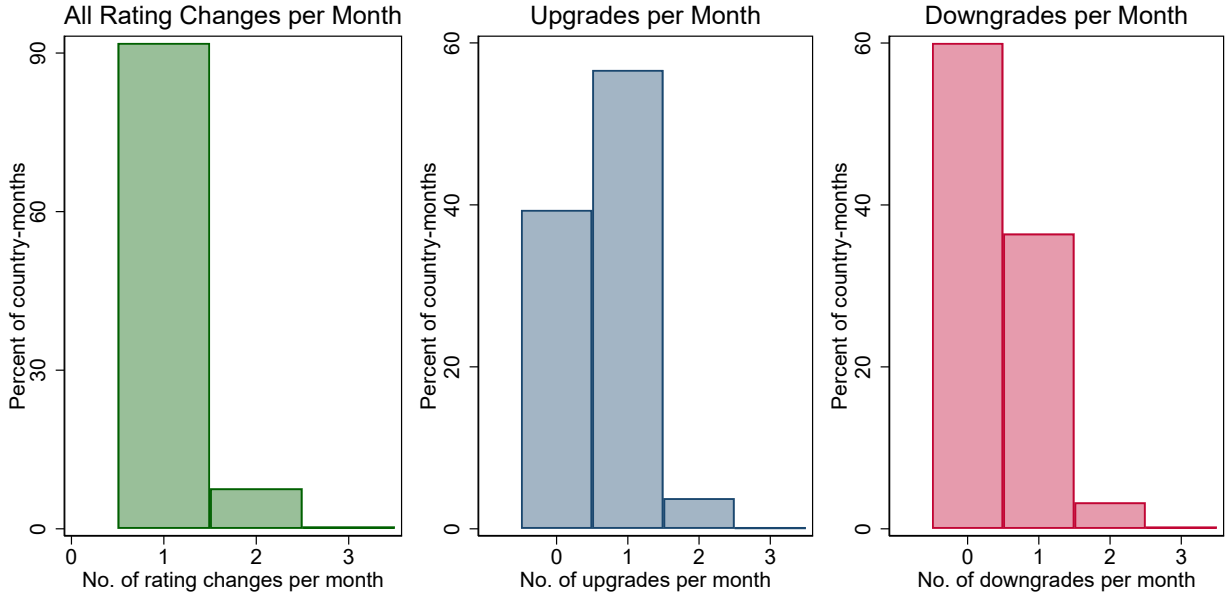
Appendix B Additional results for Section 3

Figure B.1: Bond yields around credit rating announcements, alternative trimming



Notes: The figure displays event study OLS estimates based on Equation 1 at trading day frequency, as specified in Section 3.1. The left-hand side variable is the 10-year sovereign bond yield, and the estimates correspond to an indicator variable that is activated on days of sovereign credit rating downgrades (left panel) and upgrades (right panel), respectively. The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels. The dashed lines report the point estimates of the same specification, when not winsorizing the left-hand side variable and when instead winsorizing the left-hand side variable at $\pm 1\%$ (so that the top and bottom percentile are affected).

Figure B.2: Distribution of number of rating changes per month



Notes: The figure displays the number of sovereign credit announcements by month for all rating changes (left panel) for rating upgrades (middle panel) and rating downgrades (right panel)

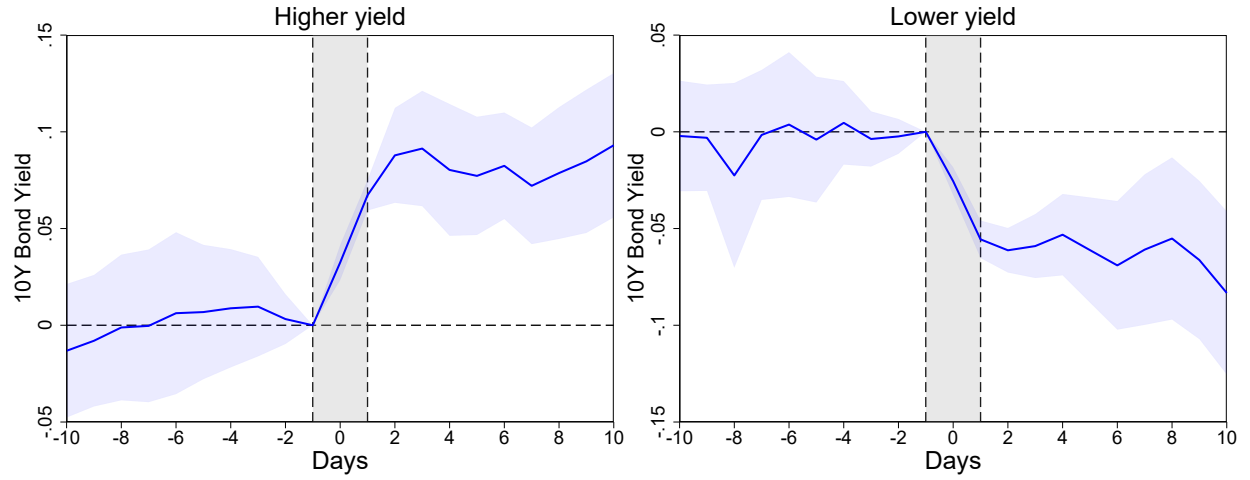
Table B.1: Summary statistics for monthly yield change surprises

	Mean	Median	Std. Dev.	Min	Max	Obs	Countries
$YCS3f_{i,t}$	-0.001	-0.002	0.110	-0.513	0.553	1162	51
$YCS3f_{i,t}^{\mathcal{D}}$	0.022	0.012	0.116	-0.442	0.450	465	47
$YCS3f_{i,t}^{\mathcal{U}}$	-0.016	-0.010	0.104	-0.513	0.553	704	51
$YCS3f_{i,t}^{\mathcal{H}}$	0.078	0.050	0.080	0.000	0.553	585	49
$YCS3f_{i,t}^{\mathcal{L}}$	-0.074	-0.049	0.081	-0.522	-0.001	630	51

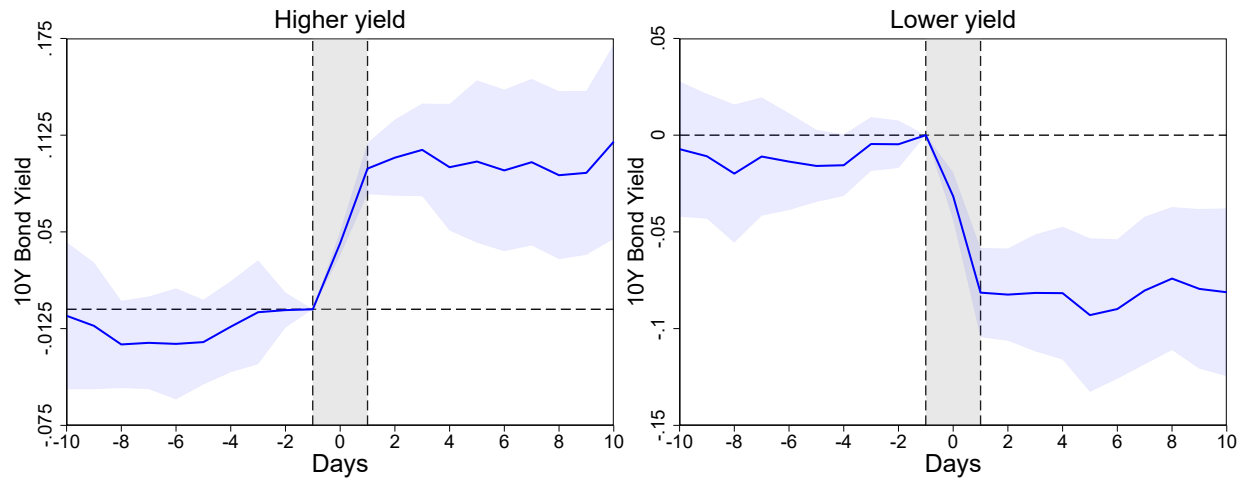
Notes: This table presents summary statistics for all yield change surprises summed to monthly frequency, as described in the text. The last column indicates the number of countries for which we have at least one non-zero surprise in the sample.

Figure B.3: Sovereign bond yields around rating change surprises, exchange rate regime

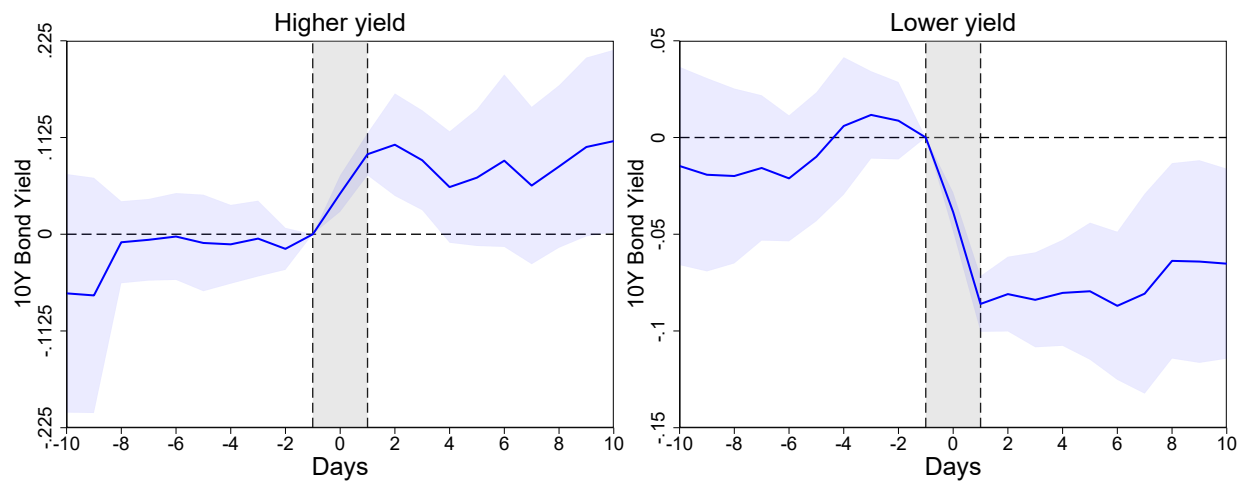
(a) Fixed exchange rate regime



(b) Floating exchange rate regime

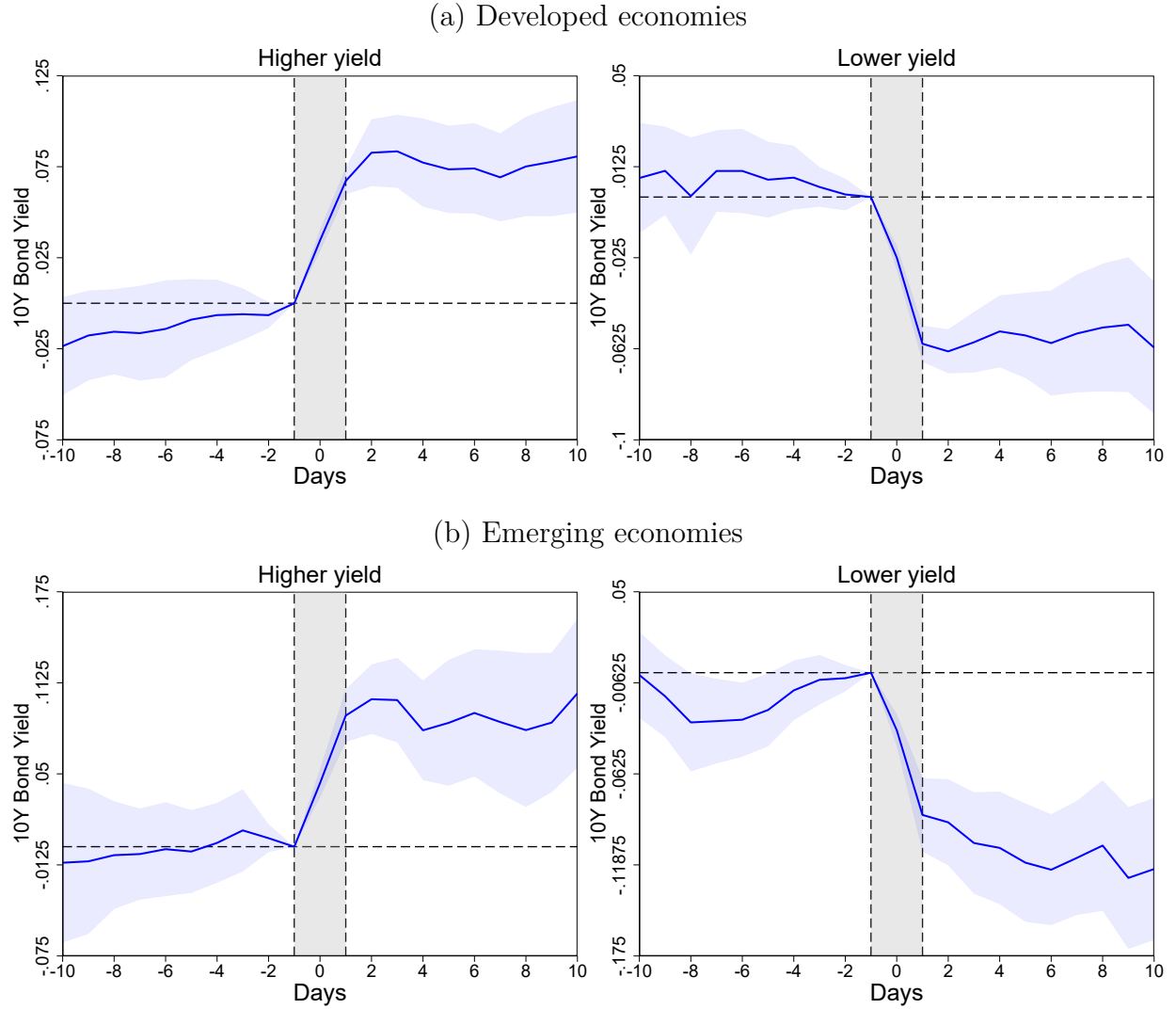


(c) Intermediate regime



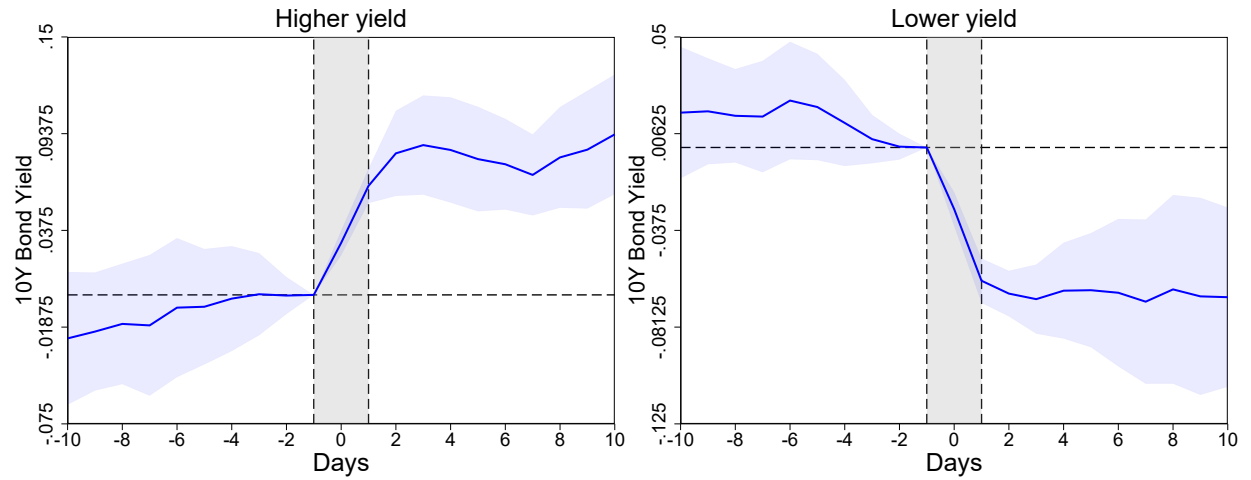
Notes: The figure displays event study OLS estimates based on Equation 1 at trading day frequency, as specified in Section 3.1. The left-hand side variable is the 10-year sovereign bond yield, and the estimates correspond to an indicator variable that is activated on days with positive (left panel) and negative yield change surprises (right panel), respectively. The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels. We further split the sample by exchange rate regimes based on Ilzetzi et al. (2019).⁶

Figure B.4: Sovereign bond yields around rating change surprises, degree of development



Notes: The figure displays event study OLS estimates based on [Equation 1](#) at trading day frequency, as specified in [Section 3.1](#). The left-hand side variable is the 10-year sovereign bond yield, and the estimates correspond to an indicator variable that is activated on days with positive (left panel) and negative yield change surprises (right panel), respectively. The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels. We further split the sample by degree of development based on [International Monetary Fund \(2023\)](#).

Figure B.5: Sovereign bond yields around rating change surprises, Euro Area membership



Notes: The figure displays event study OLS estimates based on [Equation 1](#) at trading day frequency, as specified in [Section 3.1](#). The left-hand side variable is the 10-year sovereign bond yield, and the estimates correspond to an indicator variable that is activated on days with positive (left panel) and negative yield change surprises (right panel), respectively. The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels. We further focus exclusively on Euro Area member observations.

Table B.2: Predictability of daily yield change surprises, further lags

	<i>Dependent variable: Yield-change surprise</i>				
	$YCS_{i,t}$	$YCS_{i,t}^D$	$YCS_{i,t}^U$	$YCS_{i,t}^H$	$YCS_{i,t}^L$
Stock price ($t - 1$)	-0.714* (0.401)	-0.320 (0.459)	-0.728 (0.489)	-0.203 (0.318)	-0.437 (0.276)
Stock price ($t - 2$)	0.450 (0.604)	0.229 (0.779)	0.423 (0.725)	-0.188 (0.488)	0.313 (0.505)
Stock price ($t - 3$)	0.262 (0.420)	-0.279 (0.680)	0.670 (0.478)	0.181 (0.425)	0.127 (0.345)
Stock price ($t - 4$)	0.249 (0.390)	0.647 (0.554)	-0.003 (0.435)	0.099 (0.261)	0.465 (0.324)
Stock price ($t - 5$)	-0.176 (0.368)	-0.236 (0.542)	-0.284 (0.377)	-0.041 (0.312)	0.233 (0.392)
Stock price ($t - 6$)	0.054 (0.458)	-0.373 (0.802)	0.427 (0.524)	-0.164 (0.368)	-0.159 (0.543)
Stock price ($t - 7$)	-0.315 (0.452)	0.101 (0.574)	-0.314 (0.645)	-0.036 (0.481)	-0.485 (0.497)
Stock price ($t - 8$)	0.166 (0.476)	0.038 (0.594)	-0.246 (0.631)	0.328 (0.484)	-0.178 (0.262)
Stock price ($t - 9$)	0.149 (0.368)	0.199 (0.547)	0.137 (0.518)	0.036 (0.325)	-0.102 (0.382)
Stock price ($t - 10$)	-0.123 (0.280)	0.001 (0.370)	-0.082 (0.364)	-0.024 (0.213)	0.238 (0.231)
Bond yield ($t - 1$)	0.016 (0.058)	-0.043 (0.097)	0.108 (0.080)	0.114** (0.050)	-0.055 (0.072)
Bond yield ($t - 2$)	-0.018 (0.081)	0.021 (0.115)	-0.109 (0.119)	-0.058 (0.075)	0.050 (0.094)
Bond yield ($t - 3$)	-0.010 (0.073)	-0.027 (0.087)	0.017 (0.085)	-0.063 (0.055)	-0.016 (0.065)
Bond yield ($t - 4$)	0.019 (0.075)	0.058 (0.077)	-0.052 (0.162)	0.087 (0.060)	-0.015 (0.070)
Bond yield ($t - 5$)	0.008 (0.074)	0.071 (0.057)	-0.039 (0.130)	-0.066 (0.043)	-0.011 (0.062)
Bond yield ($t - 6$)	0.010 (0.058)	-0.077 (0.096)	0.087 (0.084)	0.115* (0.061)	0.002 (0.079)
Bond yield ($t - 7$)	-0.021 (0.065)	-0.027 (0.094)	0.022 (0.128)	-0.026 (0.074)	0.019 (0.057)
Bond yield ($t - 8$)	-0.004 (0.022)	0.020 (0.031)	-0.004 (0.086)	-0.109** (0.041)	-0.031 (0.025)
Bond yield ($t - 9$)	-0.024 (0.043)	-0.055 (0.066)	0.001 (0.087)	-0.053 (0.038)	-0.023 (0.053)
Bond yield ($t - 10$)	0.025 (0.035)	0.062 (0.057)	-0.035 (0.050)	0.064* (0.032)	0.075** (0.032)
Adj. R^2	-0.015	-0.014	-0.014	0.166	0.224
Country FE	✓	✓	✓	✓	✓
Observations	1172	464	706	569	598

Notes: This table presents daily OLS regression results, as described in the text. The left-hand side variables are all non-zero yield change surprises. The regressors include a country fixed effect and daily lags of the 10-year sovereign bond yield and the main stock market index of the country under consideration. Column (1) shows all surprises pooled, columns (2) and (3) distinguish between surprises arising from credit rating downgrades ($YCS_{i,\tau}^D$) vs. credit rating upgrades ($YCS_{i,\tau}^U$), and columns (4) and (5) distinguish between positive ($YCS_{i,\tau}^H$) vs. negative surprises ($YCS_{i,\tau}^L$). Standard errors are in parentheses and clustered by country and time, and *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

Table B.3: Predictability of monthly yield change surprises

	<i>Dependent variable: Yield-change surprise</i>				
	$YCS_{i,t}$	$YCS_{i,t}^D$	$YCS_{i,t}^U$	$YCS_{i,t}^H$	$YCS_{i,t}^L$
Bond yield ($t - 1$)	-0.000 (0.001)	0.000 (0.001)	-0.001 (0.001)	0.001 (0.001)	-0.001 (0.001)
Bond yield ($t - 2$)	0.001 (0.001)	0.000 (0.001)	0.001 (0.001)	-0.000 (0.001)	0.001 (0.001)
Bond yield ($t - 3$)	-0.001 (0.001)	-0.001 (0.001)	-0.000 (0.000)	-0.001 (0.001)	-0.000 (0.000)
Ind. production ($t - 1$)	0.000 (0.000)	-0.000 (0.000)	0.000* (0.000)	0.000 (0.000)	0.000 (0.000)
Ind. production ($t - 2$)	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000* (0.000)	0.000 (0.000)
Ind. production ($t - 3$)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	-0.000* (0.000)
Unemployment ($t - 1$)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	0.000* (0.000)	-0.000 (0.000)
Unemployment ($t - 2$)	-0.000 (0.000)	-0.000 (0.000)	0.000** (0.000)	-0.000 (0.000)	0.000 (0.000)
Unemployment ($t - 3$)	-0.000 (0.000)	0.000 (0.000)	-0.000** (0.000)	-0.000 (0.000)	-0.000 (0.000)
CPI ($t - 1$)	0.000 (0.001)	0.001 (0.001)	-0.000 (0.001)	0.000 (0.001)	0.000 (0.001)
CPI ($t - 2$)	-0.000 (0.001)	0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
CPI ($t - 3$)	0.000 (0.001)	-0.000 (0.001)	0.000 (0.001)	-0.000 (0.001)	0.000 (0.001)
PPI ($t - 1$)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
PPI ($t - 2$)	-0.001 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
PPI ($t - 3$)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
LDV ($t - 1$)	-0.017 (0.027)	-0.004 (0.039)	-0.032 (0.019)	0.011 (0.023)	-0.027*** (0.005)
LDV ($t - 2$)	0.002 (0.013)	0.019 (0.023)	-0.013 (0.011)	-0.006 (0.018)	-0.014 (0.012)
LDV ($t - 3$)	0.029 (0.032)	0.021 (0.041)	-0.010 (0.039)	0.017 (0.034)	0.032 (0.031)
R^2	0.072	0.073	0.077	0.094	0.083
Country FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓
Observations	5555	5555	5555	5555	5555

Notes: This table presents monthly OLS regression results, as described in the text. The left-hand side variables are the yield change surprises, summed to monthly frequency. The regressors include country and time fixed effects and monthly lags of the 10-year sovereign bond yield, industrial production, unemployment, the consumer and producer price index (CPI and PPI, respectively), and the left-hand side variable. Column (1) shows all surprises pooled, columns (2) and (3) distinguish between surprises arising from credit rating downgrades ($YCS_{i,\tau}^D$) vs. credit rating upgrades ($YCS_{i,\tau}^U$), and columns (4) and (5) distinguish between positive ($YCS_{i,\tau}^H$) vs. negative surprises ($YCS_{i,\tau}^L$). Standard errors are in parentheses and clustered by country and time, and *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

Table B.4: Yield change surprises and monthly sovereign bond yields, only non-zero surprises

	Dependent variable: $\Delta \text{Bond Yield}_{i,t}$									
	All		Downgrades		Upgrades		Higher Yield		Lower Yield	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$YCS_{i,t}$	0.787*** (0.131)	0.781*** (0.153)								
$YCS_{i,t}^D$			0.820*** (0.193)	0.901*** (0.202)						
$YCS_{i,t}^U$					0.739*** (0.178)	0.728*** (0.214)				
$YCS_{i,t}^H$							1.398*** (0.192)	1.321*** (0.179)		
$YCS_{i,t}^L$									0.208 (0.239)	0.218 (0.258)
Partial R^2	0.066	0.067	0.058	0.071	0.070	0.071	0.097	0.092	0.002	0.003
Robust F-stat	36.2	26.1	18.0	19.8	17.2	11.6	53.3	54.6	0.8	0.7
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Lagged controls		✓		✓		✓		✓		✓
Observations	1159	1147	464	459	699	693	583	577	625	618

Notes: This table presents monthly OLS regression results based on Equation 7, as described in the text. The left-hand side variable is the monthly change in the 10-year sovereign bond yield in all columns. The regressors include the surprise of interest, a country fixed effect, and, when indicated, a vector of controls. The control vector comprises three monthly lags of the left-hand side and the surprise. Columns (1) and (2) use all surprises, columns (3) to (6) distinguish between surprises arising from credit rating downgrades ($YCS_{i,\tau}^D$) vs. credit rating upgrades ($YCS_{i,\tau}^U$), and columns (7) to (10) distinguish between positive ($YCS_{i,\tau}^H$) vs. negative surprises ($YCS_{i,\tau}^L$). Standard errors are in parentheses and clustered by country and time, and *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively. Since these estimates can be interpreted as a stylized first stage, we also present the (robust) Kleibergen-Paap F statistics. We exclude observations where the surprise takes a zero value.

Appendix C Additional results for Section 4

Table C.1: Stock market responses to sovereign credit risk, degree of development

	Dependent variable: Stock Price Surprise								
	All			Higher-Yield			Lower-Yield		
Panel A: Developed Economies									
$YCS_{i,t}$	-1.004* (0.532)	-1.021* (0.523)	-0.965* (0.538)						
$YCS_{i,t}^H$				-2.143*** (0.604)	-1.915** (0.719)	-2.214*** (0.773)			
$YCS_{i,t}^L$							-0.879 (0.832)	-2.973*** (0.752)	-2.809*** (0.791)
Within R^2	0.009	0.009	0.008	0.024	0.019	0.023	0.004	0.050	0.042
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE									
Lagged Controls									
Observations	742	726	725	370	343	343	372	350	349
Panel B: Emerging Economies									
$YCS_{i,t}$	-1.147*** (0.226)	-1.185*** (0.237)	-1.172*** (0.245)						
$YCS_{i,t}^H$				-0.951 (0.591)	-0.884 (0.605)	-0.897 (0.673)			
$YCS_{i,t}^L$							-0.516 (0.482)	-1.042 (0.956)	-1.304 (1.079)
Within R^2	0.026	0.027	0.025	0.010	0.012	0.011	0.003	0.009	0.013
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE									
Lagged Controls									
Observations	422	412	412	199	168	168	223	193	193

Notes: This table presents event study OLS estimates based on Equation 8 at trading day frequency, as specified in Section 4.1. The left-hand side variable is the change in the stock price index over a two-day window, matching the event window over which the yield change surprises are computed. The regressors include the surprise of interest, a country fixed effect, and, when indicated, quarterly time fixed effects and a vector of controls. The control vector comprises five daily lags of the asset price underlying the left-hand side and five daily lags of the 10-year sovereign bond yield. Columns (1) to (3) use all surprises, columns (4) to (6) consider positive surprises ($YCS_{i,\tau}^H$), and columns (7) to (9) consider negative surprises ($YCS_{i,\tau}^L$). Standard errors are in parentheses and clustered by country and time, and *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively. We further split the sample by degree of development based on International Monetary Fund (2023).

Table C.2: Stock market responses to sovereign credit risk, exchange rate regime

	Dependent variable: Stock Price Surprise								
	All			Higher-Yield			Lower-Yield		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Panel A: Fixed Exchange Rate Regime</i>									
$YCS_{i,t}$	-1.277** (0.600)	-1.682*** (0.426)	-1.614*** (0.482)						
$YCS_{i,t}^H$				-1.375 (0.937)	-0.772 (1.490)	-0.873 (1.779)			
$YCS_{i,t}^L$							-1.491 (1.359)	-2.731** (1.001)	-2.143* (1.188)
Within R^2	0.018	0.032	0.030	0.012	0.004	0.005	0.018	0.067	0.034
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE									
Lagged Controls									
Observations	416	399	399	219	194	194	197	173	173
<i>Panel B: Floating Exchange Rate Regime</i>									
$YCS_{i,t}$	-1.186*** (0.282)	-1.299*** (0.386)	-1.328*** (0.381)						
$YCS_{i,t}^H$				-1.467** (0.528)	-1.882** (0.685)	-1.927** (0.875)			
$YCS_{i,t}^L$							-0.927 (0.597)	-2.000*** (0.667)	-2.006* (1.108)
Within R^2	0.031	0.036	0.036	0.027	0.063	0.061	0.007	0.026	0.026
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE									
Lagged Controls									
Observations	349	334	333	168	137	137	181	155	154

Notes: This table presents event study OLS estimates based on Equation 8 at trading day frequency, as specified in Section 4.1. The left-hand side variable is the change in the stock price index over a two-day window, matching the event window over which the yield change surprises are computed. The regressors include the surprise of interest, a country fixed effect, and, when indicated, quarterly time fixed effects and a vector of controls. The control vector comprises five daily lags of the asset price underlying the left-hand side and five daily lags of the 10-year sovereign bond yield. Columns (1) to (3) use all surprises, columns (4) to (6) consider positive surprises ($YCS_{i,t}^H$), and columns (7) to (9) consider negative surprises ($YCS_{i,t}^L$). Standard errors are in parentheses and clustered by country and time, and *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively. We further split the sample by exchange rate regimes based on Ilzetzi et al. (2019).

Table C.3: Stock market responses to sovereign credit risk, no winsorizing

	Dependent variable: Stock Price Surprise								
	All			Higher-Yield			Lower-Yield		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$YCS_{i,t}$	-1.091*** (0.288)	-0.958*** (0.279)	-0.968*** (0.272)						
$YCS_{i,t}^H$				-1.599*** (0.547)	-1.822*** (0.643)	-1.577*** (0.558)			
$YCS_{i,t}^L$							-0.557 (0.545)	-1.883*** (0.622)	-1.881*** (0.655)
R^2	0.013	0.010	0.010	0.013	0.017	0.012	0.002	0.019	0.018
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE									
Lagged Controls									
Observations	1164	1154	1153	569	557	557	595	579	578

Notes: This table presents event study OLS estimates based on Equation 8 at trading day frequency, as specified in Section 4.1. The left-hand side variable is the change in the stock price index over a two-day window, matching the event window over which the yield change surprises are computed. The regressors include the surprise of interest, a country fixed effect, and, when indicated, quarterly time fixed effects and a vector of controls. The control vector comprises five daily lags of the asset price underlying the left-hand side and five daily lags of the 10-year sovereign bond yield. Columns (1) to (3) use all surprises, columns (4) to (6) consider positive surprises ($YCS_{i,t}^H$), and columns (7) to (9) consider negative surprises ($YCS_{i,t}^L$). Standard errors are in parentheses and clustered by country and time, and *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively. We do not winsorize the left-hand side variable.

Table C.4: Corporate bond price responses to sovereign credit risk, no winsorizing

	<i>Dependent variable: Corporate Bond Price Surprise</i>								
	All			Higher-Yield			Lower-Yield		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$YCS_{i,t}$	-6.212*** (1.192)	-6.869*** (1.209)	-6.390*** (1.022)						
$YCS_{i,t}^H$				-5.962*** (1.329)	-6.292*** (1.410)	-3.776*** (1.041)			
$YCS_{i,t}^L$							-6.802*** (1.092)	-4.232** (1.523)	-2.645 (1.546)
R^2	0.330	0.355	0.350	0.228	0.286	0.148	0.215	0.087	0.052
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE		✓	✓		✓	✓		✓	✓
Lagged Controls		✓	✓		✓	✓		✓	✓
Observations	232	218	218	117	90	90	115	85	85

Notes: This table presents event study OLS estimates based on Equation 8 at trading day frequency, as specified in Section 4.1. The left-hand side variable is the change in corporate bond prices over a two-day window, matching the event window over which the yield change surprises are computed. The regressors include the surprise of interest, a country fixed effect, and, when indicated, quarterly time fixed effects and a vector of controls. The control vector comprises five daily lags of the asset price underlying the left-hand side and five daily lags of the 10-year sovereign bond yield. Columns (1) to (3) use all surprises, columns (4) to (6) consider positive surprises ($YCS_{i,t}^H$), and columns (7) to (9) consider negative surprises ($YCS_{i,t}^L$). Standard errors are in parentheses and clustered by country and time, and *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively. We do not winsorize the left-hand side variable.

Table C.5: Stock market responses to sovereign credit risk, degree of development and no winsorizing

	<i>Dependent variable: Stock Price Surprise</i>								
	All			Higher-Yield			Lower-Yield		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Panel A: Developed Economies</i>									
$YCS_{i,t}$	-0.976* (0.565)	-1.034* (0.562)	-0.953 (0.573)						
$YCS_{i,t}^H$				-1.894** (0.728)	-2.146** (0.897)	-2.314** (1.097)			
$YCS_{i,t}^L$							-0.721 (0.942)	-3.139*** (0.793)	-2.949*** (0.847)
R^2	0.008	0.008	0.007	0.013	0.018	0.018	0.002	0.044	0.037
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE		✓	✓		✓	✓		✓	✓
Lagged Controls		✓	✓		✓	✓		✓	✓
Observations	742	726	725	370	343	343	372	350	349
<i>Panel B: Emerging Economies</i>									
$YCS_{i,t}$	-1.183*** (0.271)	-1.237*** (0.273)	-1.224*** (0.272)						
$YCS_{i,t}^H$				-1.336 (0.830)	-1.349 (1.018)	-1.096 (0.809)			
$YCS_{i,t}^L$							-0.403 (0.582)	-0.955 (0.996)	-1.355 (1.177)
R^2	0.024	0.026	0.024	0.013	0.019	0.011	0.001	0.006	0.011
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE		✓	✓		✓	✓		✓	✓
Lagged Controls		✓	✓		✓	✓		✓	✓
Observations	422	412	412	199	168	168	223	193	193

Notes: This table presents event study OLS estimates based on Equation 8 at trading day frequency, as specified in Section 4.1. The left-hand side variable is the change in the stock price index over a two-day window, matching the event window over which the yield change surprises are computed. The regressors include the surprise of interest, a country fixed effect, and, when indicated, quarterly time fixed effects and a vector of controls. The control vector comprises five daily lags of the asset price underlying the left-hand side and five daily lags of the 10-year sovereign bond yield. Columns (1) to (3) use all surprises, columns (4) to (6) consider positive surprises ($YCS_{i,t}^H$), and columns (7) to (9) consider negative surprises ($YCS_{i,t}^L$). Standard errors are in parentheses and clustered by country and time, and *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively. We further split the sample by degree of development based on International Monetary Fund (2023). We do not winsorize the left-hand side variable.

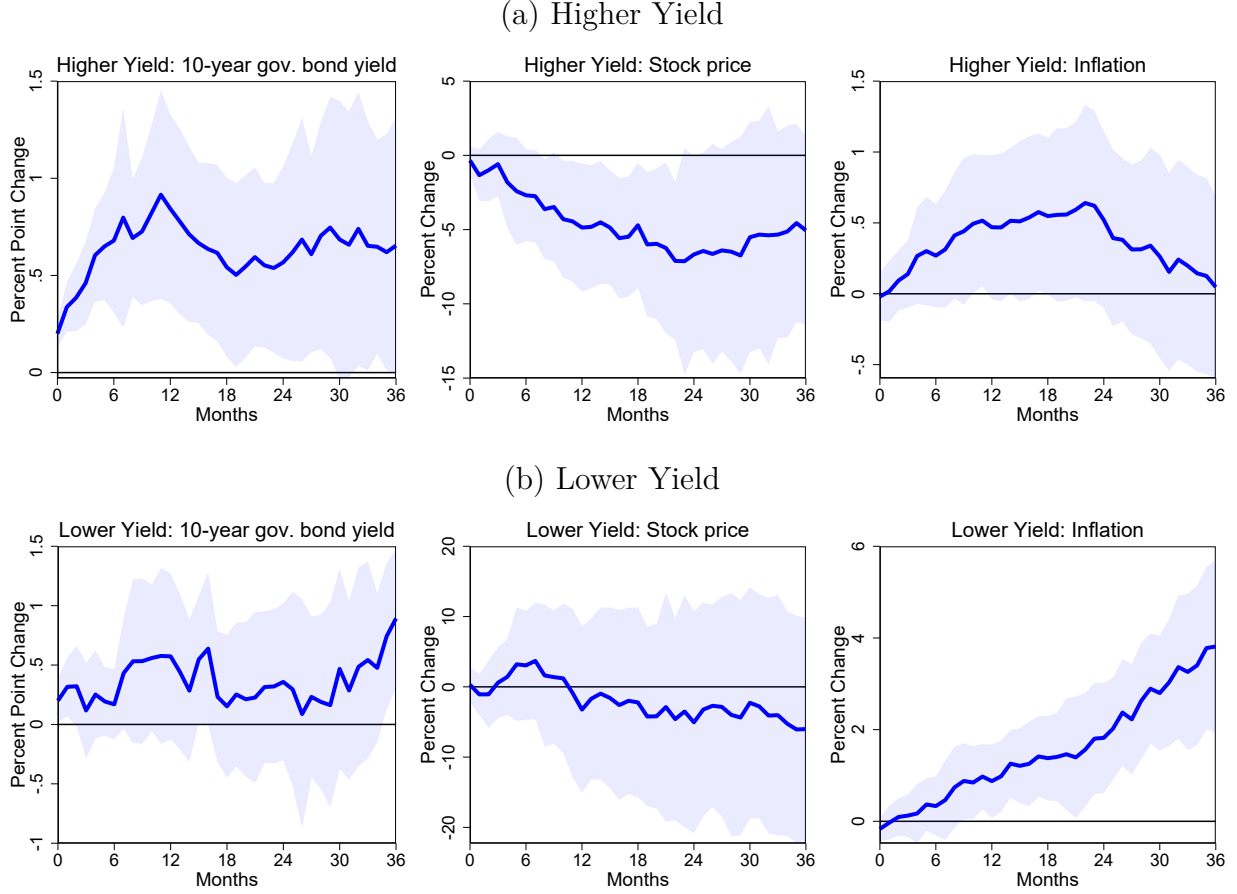
Table C.6: Stock market responses to sovereign credit risk, exchange rate regime and no winsorizing

	Dependent variable: Stock Price Surprise								
	All			Higher-Yield			Lower-Yield		
Panel A: Fixed Exchange Rate Regime									
$YCS_{i,t}$	-1.207* (0.639)	-1.684*** (0.487)	-1.620*** (0.533)						
$YCS_{i,t}^H$				-0.901 (1.144)	-0.737 (1.938)	-0.790 (2.342)			
$YCS_{i,t}^L$							-1.619 (1.379)	-2.995*** (1.019)	-2.410* (1.250)
R^2	0.014	0.028	0.026	0.004	0.003	0.003	0.017	0.071	0.038
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE		✓	✓		✓	✓		✓	✓
Lagged Controls			✓			✓			✓
Observations	416	399	399	219	194	194	197	173	173
Panel B: Floating Exchange Rate Regime									
$YCS_{i,t}$	-1.260*** (0.291)	-1.398*** (0.424)	-1.402*** (0.395)						
$YCS_{i,t}^H$				-1.971** (0.698)	-2.471** (0.954)	-2.200** (0.915)			
$YCS_{i,t}^L$							-0.964 (0.577)	-2.025*** (0.696)	-1.994 (1.174)
R^2	0.033	0.040	0.039	0.039	0.082	0.063	0.007	0.024	0.023
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE		✓	✓		✓	✓		✓	✓
Lagged Controls			✓			✓			✓
Observations	349	334	333	168	137	137	181	155	154

Notes: This table presents event study OLS estimates based on Equation 8 at trading day frequency, as specified in Section 4.1. The left-hand side variable is the change in the stock price index over a two-day window, matching the event window over which the yield change surprises are computed. The regressors include the surprise of interest, a country fixed effect, and, when indicated, quarterly time fixed effects and a vector of controls. The control vector comprises five daily lags of the asset price underlying the left-hand side and five daily lags of the 10-year sovereign bond yield. Columns (1) to (3) use all surprises, columns (4) to (6) consider positive surprises ($YCS_{i,\tau}^H$), and columns (7) to (9) consider negative surprises ($YCS_{i,\tau}^L$). Standard errors are in parentheses and clustered by country and time, and *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively. We further split the sample by exchange rate regimes based on Ilzetzi et al. (2019). We do not winsorize the left-hand side variable.

Appendix D Additional results for Section 5

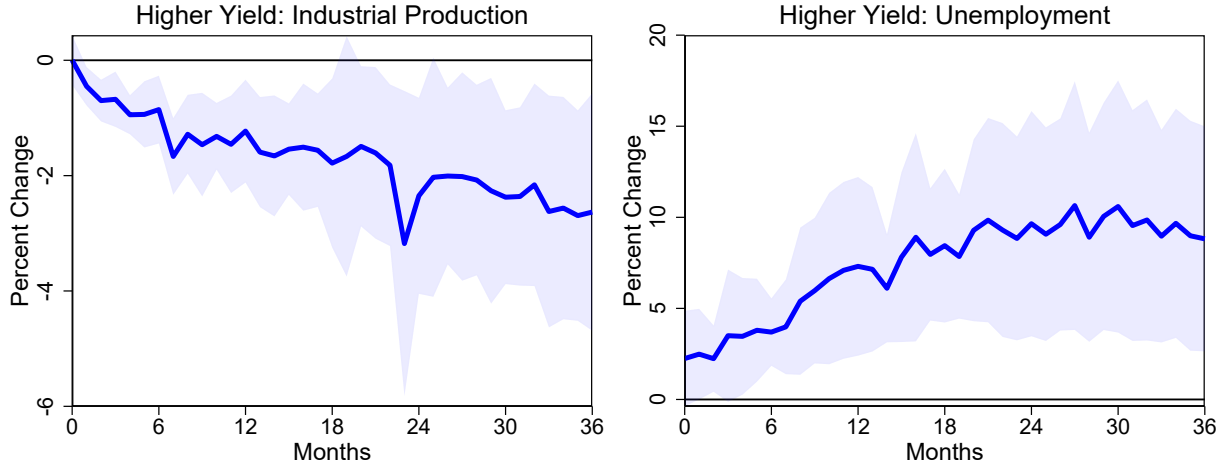
Figure D.1: Responses of prices to sovereign credit risk shocks, sign of the surprise



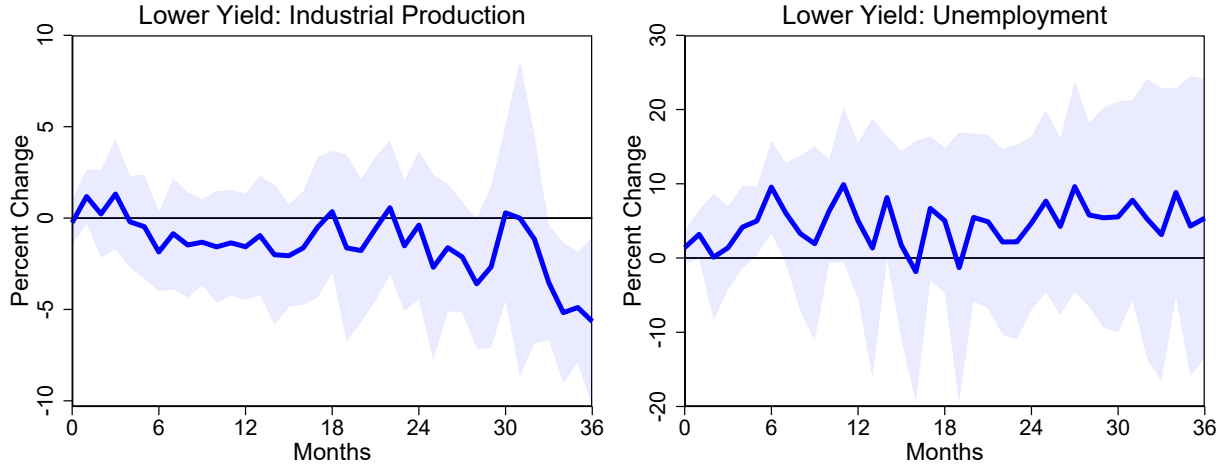
Notes: The figure displays monthly OLS local projection estimates based on [Equation 9](#), as specified in [Section 5.1](#). The left-hand-side variables are 10-year sovereign bond yields, stock price indices, and consumer prices and the regressor of interest is the yield change surprise. All estimates correspond to an initial yield increase of 0.2 percentage points. We further include country and time fixed effects, three lags of the left-hand side variable and the bond yield as controls. The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels. We further split the surprise series to distinguish between positive (higher yield) and negative (lower yield) surprises.

Figure D.2: Responses of real activity to sovereign credit risk shocks, sign of the surprise

(a) Higher Yield

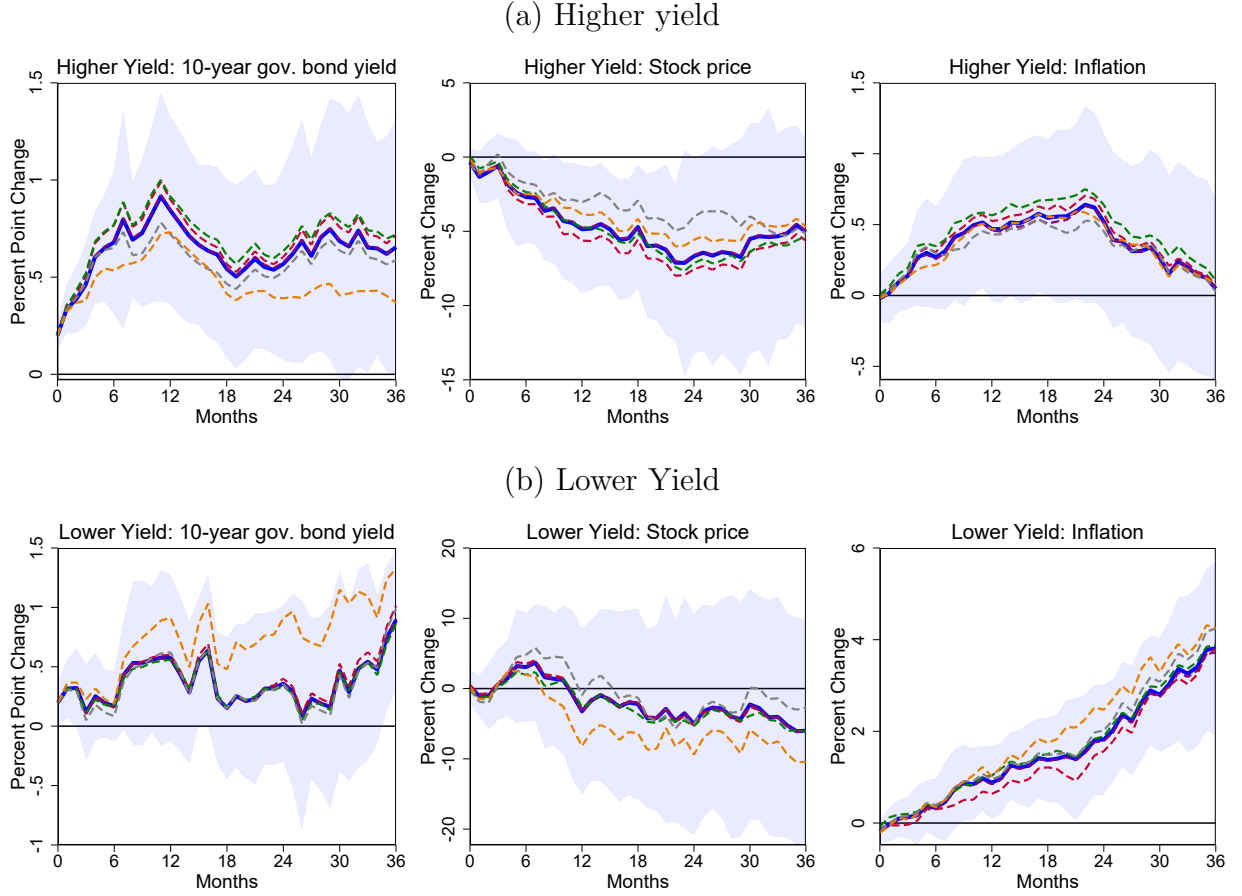


(b) Lower Yield



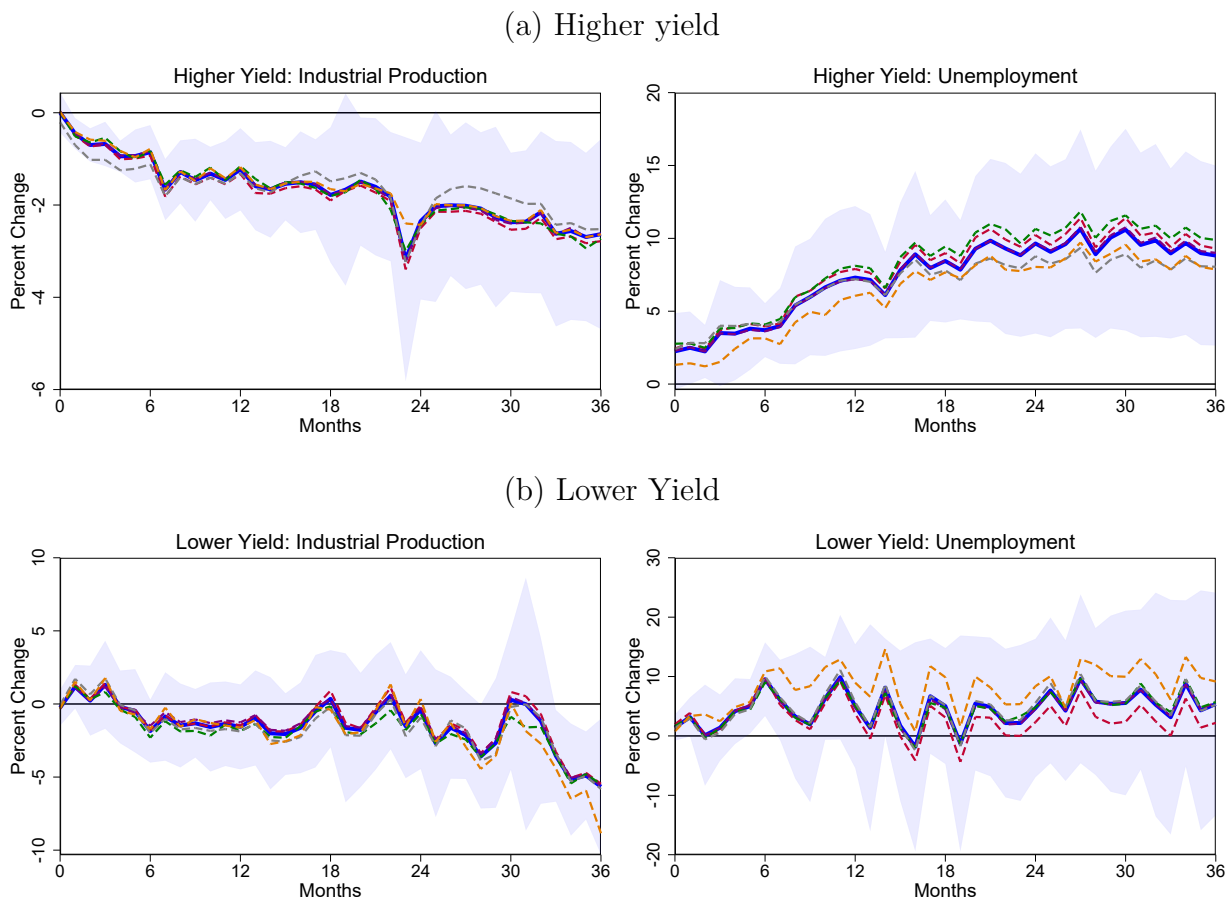
Notes: The figure displays monthly OLS local projection estimates based on [Equation 9](#), as specified in [Section 5.1](#). The left-hand-side variables are industrial production and unemployment and the regressor of interest is the yield change surprise. All estimates correspond to an initial yield increase of 0.2 percentage points. We further include country and time fixed effects, three lags of the left-hand side variable and the bond yield as controls. The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels. We further split the surprise series to distinguish between positive (higher yield) and negative (lower yield) surprises.

Figure D.3: Responses of prices to sovereign credit risk shocks, robustness and sign of the surprise



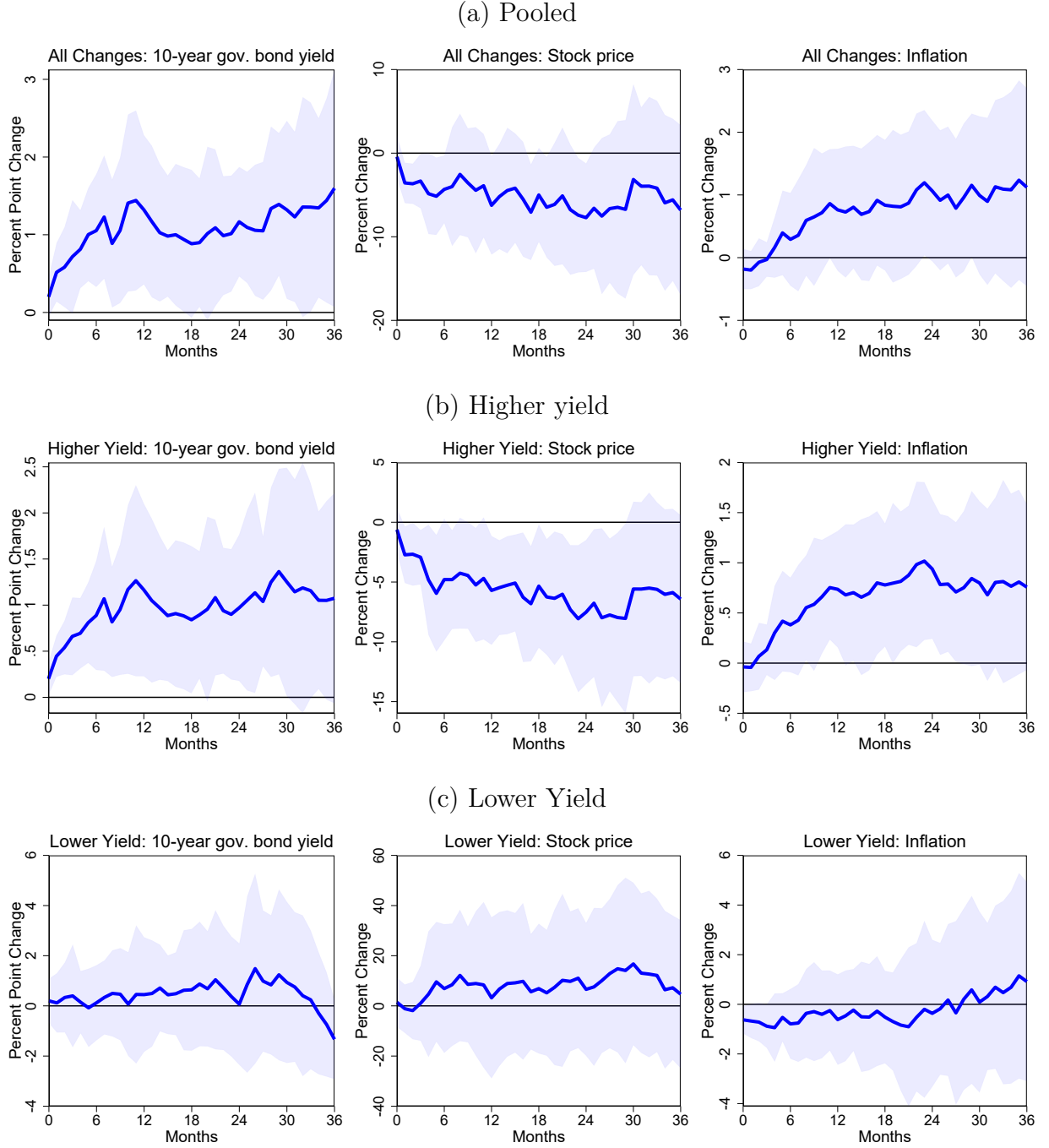
Notes: The figure displays monthly OLS local projection estimates based on Equation 9, as specified in Section 5.1. The left-hand-side variables are 10-year sovereign bond yields, stock price indices, and consumer prices and the regressor of interest is the yield change surprise. All estimates correspond to an initial yield increase of 0.2 percentage points. The blue lines and shaded areas reproduce the baseline specification, which includes country and quarterly time fixed effects and three lags of the left-hand side variable and the bond yield as controls. The remaining lines show alternative specifications: extending the baseline controls from three to six lags (red); adding three lags of the yield change surprise to the baseline controls (purple); replacing quarterly time fixed effects with monthly fixed effects (green); replacing quarterly time fixed effects with yearly fixed effects (grey); and winsorizing the left-hand side variable at 0.5% (orange). The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels. We further split the surprise series to distinguish between positive (higher yield) and negative (lower yield) surprises.

Figure D.4: Responses of real activity to sovereign credit risk shocks, robustness and sign of the surprise



Notes: The figure displays monthly OLS local projection estimates based on [Equation 9](#), as specified in [Section 5.1](#). The left-hand-side variables are industrial production and unemployment and the regressor of interest is the yield change surprise. All estimates correspond to an initial yield increase of 0.2 percentage points. The blue lines and shaded areas reproduce the baseline specification, which includes country and quarterly time fixed effects and three lags of the left-hand side variable and the bond yield as controls. The remaining lines show alternative specifications: extending the baseline controls from three to six lags (red); adding three lags of the yield change surprise to the baseline controls (purple); replacing quarterly time fixed effects with monthly fixed effects (green); replacing quarterly time fixed effects with yearly fixed effects (grey); and winsorizing the left-hand side variable at 0.5% (orange). The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels. We further split the surprise series to distinguish between positive (higher yield) and negative (lower yield) surprises.

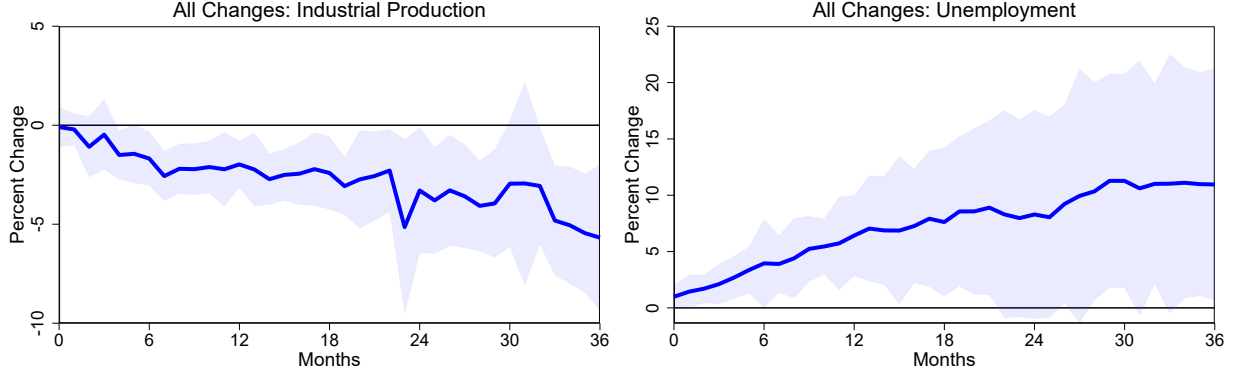
Figure D.5: Financial outcomes, three lags of all outcomes included (implicit sample fix)



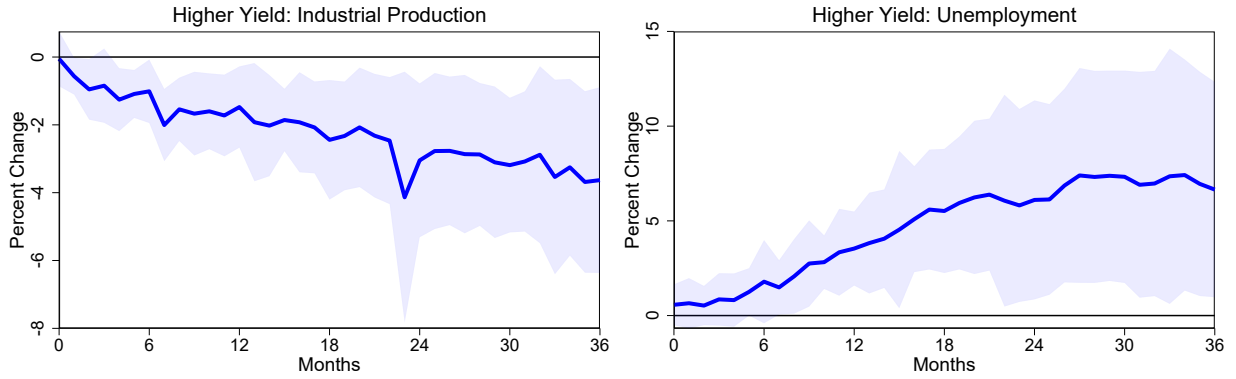
Notes: The figure displays monthly OLS local projection estimates based on [Equation 9](#), as specified in [Section 5.1](#). The left-hand-side variables are 10-year sovereign bond yields, stock price indices, and consumer and the regressor of interest is the yield change surprise. All estimates correspond to an initial yield increase of 0.2 percentage points. We further include country and quarterly time fixed effects and three lags of all macro-outcomes (10-year sovereign bond yields, stock price indices, consumer prices, industrial production, and unemployment) as controls. Note that this restricts the sample to observations for which all outcomes are jointly available. The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels. We further split the surprise series to distinguish between positive (higher yield) and negative (lower yield) surprises.

Figure D.6: Real outcomes, three lags of all outcomes included (implicit sample fix)

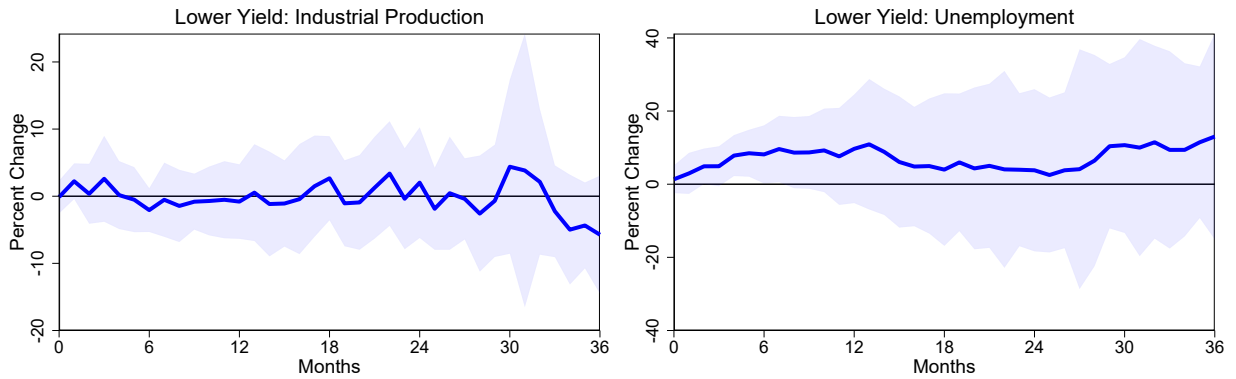
(a) Pooled



(b) Higher yield



(c) Lower Yield



Notes: The figure displays monthly OLS local projection estimates based on [Equation 9](#), as specified in [Section 5.1](#). The left-hand-side variables are industrial production and unemployment and the regressor of interest is the yield change surprise. All estimates correspond to an initial yield increase of 0.2 percentage points. We further include country and quarterly time fixed effects and three lags of all macro-outcomes (10-year sovereign bond yields, stock price indices, consumer prices, industrial production, and unemployment) as controls. Note that this restricts the sample to observations for which all outcomes are jointly available. The blue-shaded areas indicate 95% confidence bands based on standard errors clustered at the country and time levels. We further split the surprise series to distinguish between positive (higher yield) and negative (lower yield) surprises.